

The Fable Computer, Part III: A Ternary Language Model on the Regenerative Graphene-Plasmon Fabric

Embedding Ternary Bonsai 2 27B as a streaming matrix-multiply coprocessor — a digital lane on the Part-I logic fabric, bit-exact as a netlist, a cold analog counter tree on the Part-II interference fabric, and the one measurement that decides both

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Authorship and Status Disclaimer. This manuscript extends the author’s design studies [1, 2] (“Part I”, “Part II” throughout). Generation of content involved the use of an AI, Anthropic’s Claude, at the author’s direction: the two design studies folded into Sections 3–7 were executed in a Claude Code session (model Claude Opus 5.5) and each was re-checked by an independent adversarial verification agent before its second revision; the manuscript itself was drafted on Claude Opus 5.5 under orchestration by Claude Fable 5.1 and revised after an AI proofreading pass. The author declares a lack of formal expertise in machine-learning accelerator architecture and in graphene-plasmonic device physics, and the document has not received independent expert review. All physical claims are reduced-order model results in the sense of Parts I and II (“in-model”: established within the released chains and the named extensions, under stated assumptions, not measured) and must be treated as a starting point for discussion, not as peer-reviewed engineering guidance.

Abstract

A ternary-weight language model is a natural workload for the regenerative graphene-plasmon fabric of Parts I and II: its inner loop is “add, subtract or skip an 8-bit number”, and the fabric’s primitive is a threshold cell that adds and restores. Ternary Bonsai 2 27B [3–5] has 27 B weights in $\{-1, 0, +1\}$; one decoded token takes 25.6 G ternary multiply-accumulates (MACs) and streams 5.67 GB of weights (external). The embedding is a streaming coprocessor: a conventional (CMOS) host keeps every stateful and floating-point task, and fabric lanes — fixed datapaths of cells — each compute one 32-wide block sum of 8-bit activations and ternary weights per 14.5-ps slot, assuming the llama.cpp family’s Q8_0 activation blocks. The digital lane is bit-exact (demonstrated (runnable) on 1.70 M Bonsai-shaped block sums; a netlist check, not physics) and, timed for latch-free operation, takes 8,169–8,455 cells, 0.22–0.23 mm² and 64–86 mW, or 29–39 fJ per MAC, at 300 K at the hydrodynamic tier (in-model; the kinetic correction to the dissipation is not evaluated at 300 K). Two, three and seven lanes keep pace with 0.5, 1 and 3 TB/s (88, 176 and 529 tok/s at batch 1) in at most about 1.7 mm² and 0.64 W (in-model); the system stays memory- and I/O-bound. Against the model’s own activation-quantization error, a lane may err at most 6.3×10^{-9} per cell decision (in-model; an upper bound). Only the released booking of that error passes warm: Part I’s worst-path budget lowers the pass temperature to 144 K for its –3 dB split alone and to 114 K through a –1 dB junction (colder at –3 dB), and the operand launch must be 28–33 dB quiet against the record’s coded 20 dB (in-model). On the Part-II interference fabric, a three-input interference decoded by three comparators (QFA-3) can replace 184, 154, 65 and 15 of the Wallace tree’s 236 full adders at 4, 20, 77 and 300 K (in-model; five rework items open), saving at most about a third of the cells ($\leq 32\%$) even if every counter were analog; a 77 K lane wins only with a cooler at $\geq 40\%$ of Carnot, and 20 K and 4 K lanes lose even at Carnot (in-model; cold-cell dissipation uncertified). Both routes end at one open measurement, the loaded decision error of a cell versus temperature (bench gate DG1); the digital lane is recommended for Bonsai, and QFA-3 is kept as a test tile. Of the levers (in-model, on synthetic activations), 4-bit activations would raise the worst-path (A2–A3) pass temperatures from 114–144 K to 168–213 K, and a threshold full adder would halve lane logic. Nothing has been built; the gain cell is experimentally unproven.

1. Introduction: a model that only adds, and a fabric that only adds

A large language model generates text one token at a time. Each decode step pushes one activation vector through every layer, and almost all of its arithmetic is matrix-vector products: every weight is read once, multiplied by one activation and added into an output. When one user is served at a time (batch 1), each weight fetched from memory is used once, so the processor waits on memory, and the decode rate is the memory bandwidth divided by the bytes of weights read per token — the memory-bound corner of the roofline model [6, 7] (external).

Ternary models change the arithmetic, not that bound. The BitNet b1.58 line of work [8], which grew out of the binary-weight BitNet [9], trains networks whose weights are -1 , 0 or $+1$, with one floating-point scale per tensor (per group of 128 in Bonsai), while activations stay at 8 or 16 bits: a multiply becomes “add”, “subtract” or “skip”, and a weight costs about 1.7 bits instead of 16. Ternary Bonsai 2 27B [3–5], derived from Qwen3.8-27B [10], is an openly released instance at a scale where memory bandwidth dominates; it keeps 98.2 % of its parent’s benchmark average (external; Section 3).

The Fable Computer is a machine design on a graphene-plasmon fabric (Section 2). In Part I [1] every gate is a graphene cavity about half a micrometre long, biased by a DC current just below self-oscillation, so that it amplifies a terahertz plasmon pulse and clamps it against a saturating level, the rail; logic is a threshold on the sum of input fields. The fabric runs wave-pipelined — operations follow each other through the cells one clock slot apart, with no latches — and stores nothing. Part II [2] showed that below the rail the same fabric is a quantum-limited analog computer whose interference the same threshold cells decode to one or two bits. The gain cell itself has never been demonstrated.

The workload and the fabric fit. At batch 1 the weights stream from DRAM on every token anyway, so the fabric’s lack of storage forfeits no weight reuse that a batch-1 GPU would have had; what it does cost is that every activation bit must be relaunched in the slot that uses it, which makes the operand lines the hardest engineering in the design (Section 4.6). The fabric cannot decode faster than the memory allows, so the questions are what maps onto it exactly, whether that piece belongs on Part I’s restoring logic or on Part II’s interference, and which unmade measurement decides whether it works.

In brief: a sum of 32 products of 8-bit activations and ternary weights per slot — the Q8_0 activation block of the llama.cpp family [11], adopted here as an assumption — fits on a Part-I lane, a fixed datapath of cells, that is bit-exact as a netlist and needs about 8,200–8,500 cells; two to seven lanes keep pace with 0.5–3 TB/s (Section 4). Whether the lane is also correct depends on how often a cell decision flips: the record’s pieces admit five bookings, spanning seven orders of magnitude in lane error power at room temperature (lane signal-to-noise ratio, SNR, from 64.9 to -4.0 dB; Table T7), and only the released one passes (Section 5). Part II’s interference counters help only when cold, saving at most about a third of the cells, while the cooler costs more unless it is a good one (Section 6). Both routes end at bench gate DG1, a pre-registered pass/fail measurement of a loaded cell’s decision error versus temperature (Section 7).

Section 2 summarizes Parts I and II (Tables T1, T2); Sections 3–8 develop the design; Section 9 states what is demonstrated, in-model, external and open and lists the limitations; Section 10 concludes; the appendices document the listings and the symbols. The labels of Parts I and II are kept: **demonstrated** — executed against the released, unedited chains (for a netlist identity, “demonstrated (runnable); a netlist check, not physics”); **in-model** — established within the record’s models plus this paper’s named extensions, under stated assumptions, not measured; **open**; and **external**, for facts about Bonsai, GPUs and memory. “The record” is the two released model chains plus the community notes promoted into the project repository [12], cited by date.

2. The fabric in brief (what a reader needs from Parts I and II)

This section is for readers new to Parts I and II. Every fabric term is defined at first use and collected in Table T1; Section 2.5 states what Part III keeps, changes and considered instead.

2.1 One cell, every gate (Part I)

The cell is a Dyakonov–Shur (DS) cell: a 576-nm cavity of graphene encapsulated in hexagonal boron nitride, at a carrier density $n = 10^{12} \text{ cm}^{-2}$ under a metal gate, resonant at 1 THz for the gate-screened plasmon. A DC drift current flows through it; Dyakonov and Shur showed that such a channel, with electrically different ends, self-oscillates above a threshold drift. Part I biases the cell at 0.7 of that threshold (a tracked ratio, recalibrated in situ), where it amplifies instead: an injected pulse recirculates and leaves stronger. This regenerative gain is +9.7 dB for a continuous wave (CW) and about +8 dB per pulse streaming, in-model at the hydrodynamic tier (the record’s fluid model of the electrons), with a signed −0.95 dB per cell from the finer kinetic tier at 353 K still to be certified, plus a separate, still-open ± 1 dB allowance for the clamped contacts that could be larger for a diffuse contact, one that randomizes the momentum of the electrons it reflects, where a specular one reflects them mirror-like (Part I §7.3, §7.5; note 2026-07-13). The pulse saturates against the rail, so every cell restores logic levels.

Logic uses the same transfer: a cell fires when the summed input fields on its gate cross its threshold — a high threshold makes AND, a low one OR, and a depleting coupling (pushing the cell below threshold when the input is high) makes NOT. The steepness is the sharpness $k = \beta \cdot A_{\text{swing}} / \Delta n$ (coupling times logic swing over threshold width); the design rule $k \geq 8$ gives a static noise margin — the input disturbance a restoring cell rejects — of 0.256 of the swing (the released value, note 2026-07-21; Part I §5 quotes ≈ 0.2). There is no multiplier and no native plasmonic storage: “compute in plasmons, store as charge” (Part I §6.4). Persistent bits live as gate charge in conventional electronics, and Part I allows only small fast registers of pumped bistable cells or recirculating loops, which the released chains do not model; this Part uses none, so its lane has no latch.

Part I’s half adder is an OR, a carry AND, a NOT, a delay buffer (DLY) and an output AND — five active cells of depth 3 — plus two optional carry-alignment buffers: seven biased cells (Part I §6.5). The DLY cell only equalizes path lengths so that both inputs of the output AND arrive in the same slot, a discipline Section 4.3 generalizes. Part I composes a full adder from “two half adders and an OR”: $2 \times 7 + 1 = 15$ biased cells of depth 7, this Part’s reference.

2.2 Timing, fan-out, stations

The clock is optical: a microcomb-photomixing chain launches phase-stable pulse trains at the fabric edge (Part I §3, §6.2). A launch is the injection of a clock or data pulse; a slot is one clock period. Operation is wave-pipelined [13], so timing rests on balanced path lengths, and re-sync stations — DS cells used as clocked ANDs, fed by optical clock taps — periodically re-time pulses and flush noise. Part I wrote “every 10–24 gates”; the released spacing law gives 10.0 cells at a cell noise figure (NF, the factor by which a cell degrades the signal-to-noise ratio) of 4 dB and 13.3 cells at the 2.77-dB NF floor, and no documented margin convention reaches 24 (note 2026-08-25 §5 item 5; digital_lane_bonsai.py §0). This Part uses 10–13.

Fan-out F costs 3 dB per branch of a Y-split plus junction loss; a junction is a gate-defined step between cells, budgeted at −1 to −3 dB (Part I §6.3, §9.4). Only $F = 2$ closes: note 2026-07-21 gives an $F = 2$ reserve of +2.01 dB, an effective sharpness $k_{\text{eff}} = 8 \cdot F^{(-1/2)} = 5.66$ (static noise margin 0.221) and a disorder ceiling $F \leq 2.56$, the largest fan-out at which k_{eff} stays above the regeneration floor $k = 5$ (in-model). Wider fan-out uses trees of $F = 2$ buffers.

The slot rate has moved since Part I's 4-ps (0.25 THz) baseline. Because a regenerative cavity rings for a few picoseconds after a pulse, each slot must be flushed, and $F = 2$ logic then closes only near 0.1 THz (note 2026-07-23); with a physical launch confined to its slot and gated by the data, the $F = 2$ frontier is 69 GHz [63, 74] (central value and band), adjudicated at the 353 K bias (note 2026-07-31; in-model). Three notes of 2026-08-01 qualify it. The gain-requirement check confirms the window; the reissued bench gate G1, a proposal, keeps its CW existence test (G1a) and adds a gated-drive knee report (G1b); and at the ratio bias the bench will use ($0.7 \times$ the measured threshold) the '0'-slot floor — the largest output in a '0' slot, referred to the mean '1' — binds: under the strict (11.8-dB) floor specification no gated slot rate passes robustly; under a round ≥ 10 dB specification, 67 GHz is robust and 71 GHz marginal. Note 2026-08-02 (revised and re-submitted) finds that the released chain cannot fix the floor specification, which must be imported, and at its fitted convergence order resolves 71 GHz to a 0.31-dB fail of the round specification (in-model). The frontier, 69 GHz [63, 74], is neither confirmed nor moved (open). This paper uses 69 GHz (a 14.5-ps slot); every lane count, tok/s per lane and line rate below scales with the slot rate.

The released chain stores a cell dissipation of 12.67 μW at the 353 K (80 °C) cap, with an upper bound of 16.64 μW (1.267 and 1.664 kW/cm² over 10^{-8} cm², hydrodynamic tier). Scaled to 300 K as τ^{-3} at the tracked bias (τ the plasmon lifetime; extension [D3] — bracketed labels [D...], [E...] and [V...] name this paper's in-model extensions, defined in Table TA3) it is 7.78–10.22 μW ; kinetic-corrected at 353 K it is 15.9–16.7 μW (digital_lane_bonsai.py §0). A cell is charged about 11 sites of Part I's gate lattice (27.4 μm^2 , the half-adder block's $12 \times 16 \mu\text{m}$ over its 7 biased cells; Part I Table 5), of which its active area is 1 μm^2 , so cells fill 3.6 % of a lane. The chip may rise 55 K above ambient; with the kinetic correction Part I's margin at a one-third fill (the area fraction under biased cells) is about 4 % (Part I §8.2).

2.3 The quantum view (Part II)

Part II counts pulses in quanta. One plasmon of the 1-THz cell mode is a 0.16 % density swing; the knee (the 1-dB compression point, where the linear window ends) is ≈ 38 quanta at the launch plane (band 22–51; a reference plane is the point along a pulse's path at which an amplitude is quoted), and the rail $\approx 3,833$ quanta, anchored at no plane (Part II §3.1; note 2026-07-17) — classical solver amplitudes converted to quanta. With $T_Q = \hbar\omega_0/k_B \approx 48$ K, the mode carries $\bar{n} = 5.8$ thermal quanta at 300 K and about one at 77 K. Part II excludes quantum logic on this cell and builds a quantum-limited analog computer under two rules: gain only after a decision, and a quiet launch.

Its unit QMAC-1 interferes two pulses of N_{op} quanta in one junction and decodes the sum with two comparator cells — Part I's half adder re-programmed, except that the comparators need $k \geq 16$, a redesign (Part II §5). Its 2-bit decode errs 0.14 per shot at 300 K, its 1-bit decode 3.9×10^{-3} , and a rail-restored digital decision 3.6×10^{-11} (Part II Table Q3). This paper calls that last figure booking A0, a booking being a stated convention for turning a decision's swing and noise into an error rate (Section 5.3).

Three Part-II quantities are open. The launch: Part I budgets a 20-dB launch SNR and Part II codes it as 10 % amplitude jitter — two different quantities (note 2026-08-25) — at which the 2-bit decode cannot be cooled below a few times 10^{-3} (Part II §4.2). N_{op} : the default 400 quanta per input is called "launch-budget-capped", a cap the record never derives. And c, the comparator's rms trim offset as a fraction of its transition width, is coded at $1/\sqrt{12} = 0.289$ and derived nowhere (note 2026-07-15).

2.4 What is demonstrated

Nothing physical. No DS cell has been built and biased as a regenerative amplifier; Part I's bench gates G1–G5 (G1: single-cell net gain at ≤ 80 °C) and Part II's QG1–QG5 are unrun, and the one current-driven

graphene terahertz amplification reported as of Part I (June 2026) is attributed by its authors to another mechanism (Part I §2). The record is the two released chains (fable-model-chain/, fable-model-quantum/), executed unedited, and the promoted notes [12]; every physical number below inherits the cell's in-model status.

Part I also places a data gate on this Part: a near-model sub-threshold cavity linewidth on a specular-contact cell, measured with a contact-preparation contrast, a temperature sweep and a drift sweep, “gates an architecture-led Part III on data” (Part I §10), because the open ± 1 dB contact allowance dominates the per-cell gain budget behind the $F = 2$ reserve and the bookings of Section 5 (Part I §7.5; notes 2026-07-18). That measurement is unmade too; it stands in Table T10 as L0, after G1.

Term	Meaning	Value at the operating point	Where
DS cell	576-nm gated-graphene cavity with DC drift, biased below self-oscillation; every gate is one	$n = 10^{12} \text{ cm}^{-2}$, 1 THz, bias $M/M_{\text{th}} = 0.7$ (tracked)	Part I §5
Regenerative gain	multi-pass amplification of an injected pulse in a sub-threshold cell	+9.7 dB CW, $\approx +8$ dB per pulse streaming; kinetic -0.95 dB/cell at 353 K	Part I §7.3, §7.5
Rail	saturation level that re-clamps a ‘1’ at every cell	$\approx 10\%$ density swing $\approx 3,833$ quanta, anchored at no plane	Part I §5; Part II §3.1; note 07-17
Knee	1-dB gain-compression point; end of the linear window	$\approx 1\%$ swing ≈ 38 quanta at the launch plane (22–51)	Part II §3.1; note 07-17
Threshold logic	AND/OR = threshold on summed input fields; NOT = depleting coupling	high threshold AND, low threshold OR	Part I §5
Sharpness k	threshold steepness $\beta \cdot A_{\text{swing}}/\Delta n$; sets the static noise margin (NM)	$k \geq 8$: NM 0.256; $k_{\text{eff}} = 5.66$ at $F = 2$: NM 0.221	Part I §5; note 07-21
Fan-out F	cells driven by one output through Y-splits (-3 dB per branch)	$F = 2$ (reserve +2.01 dB; ceiling 2.56)	Part I §6.3; note 07-21
Junction	gate-defined step between adjacent cells	-1 to -3 dB	Part I §9.4
Slot	one clock period; one operation per pipeline per slot	14.5 ps at 69 GHz [63, 74]; open in the bench convention	notes 07-23, 07-31, 08-01, 08-02
Launch	injection of a clock or data pulse at the fabric edge	coded 20 dB = 10 % amplitude jitter	Part I §6.2; Part II §4.2; note 08-25
Wave pipelining	operations one slot apart with no latches; timing by path balance [13]	—	Part I §6.5
Re-sync station	clocked AND fed by an optical clock tap; re-times and flushes noise	every 10–13 cells (NF 4 dB to the 2.77-dB floor)	Part I §6.2, §9.6; note 08-25
Quantum, N	one plasmon of the cell mode: a 0.16 % density swing	$\hbar\omega_0 = 4.14$ meV; $T_Q = 48$ K; $\bar{n} = 5.8$ (300 K)	Part II §3
N _{op}	quanta per input pulse at logic 1 in an analog tile	400 registered; 2,600 in this paper's REF	Part II §5; §6.3
QMAC-1	two-input interference decoded to 2 bits by $k \geq 16$ comparators	2-bit error 0.14 (300 K), 3.1×10^{-3} (77 K)	Part II §5–6
c	comparator rms trim offset, fraction of its transition width	coded $1/\sqrt{12} = 0.289$; open	Part II §5; note 07-15
P1, P2	rail reference planes: the pre-loss launch sum; the combiner output	both carried; the record anchors the rail at neither	§6.2–6.3
REF, CONS, PERM	the analog tile's three conventions: reference (rail at P2, relaunched inputs, $c = 1/\sqrt{12}$); conservative (rail at P1); permissive (quiet launch, $c = 0.02$)	N_{op} 2,600 / 1,278 / 1,822 at 4 K	§6.3
Booking	convention turning a decision's swing and noise into an error rate	A0 (released) ... A3, B	§5.3
Ternary g128	weights in $\{-1, 0, +1\}$, one FP16 scale per 128	1.72 bits per weight	[3]
Q8_0	llama.cpp activation block: 32 int8 values + one FP16 scale	dot-product SNR 45.5 / 42.5 dB (Gaussian / Student-t)	[11]; §5.1
PTQ1_0, PQ2_0	PrismML's GGUF packings: dense trits; one trit per 2-bit slot	1.75 bits per weight, 5.95 GB; 7.21 GB (files on disk)	[3]

Term	Meaning	Value at the operating point	Where
3:2 counter (full adder)	adds three equal-weight bits into sum and double-weight carry	15 cells, depth 7 (2 half adders + OR)	Part I §6.5; §4.1–4.2
Wallace tree	column reduction by layers of 3:2 counters [14, 15]	236 counters in 8 stages	§4.2
Block sum, lane	32-wide int8 × ternary sum per slot; the datapath computing it	$0 \leq U \leq 8,160$; 8,169–8,455 cells	§4.1–4.3
Yardstick, pass line	the runtime’s own Q8_0 error; lane error power may not exceed it	42.5 dB; 6.3×10^{-9} per decision	§5.1–5.2
QFA-3	three-input interference decoded by three comparators: an analog full adder	1.3×10^{-5} at 4 K (REF)	§6.2–6.3

Table T1. Glossary. Each fabric term is defined at first use in the text; values are quoted at the operating point (1 THz, $n = 10^{12} \text{ cm}^{-2}$, $M/M_{th} = 0.7$, $F = 2$, 69-GHz slot) with the temperature where it matters. “§” without a Part refers to this paper.

2.5 What Part III keeps, changes and considered instead

When Part III adopts a construct from Part I or II, it keeps it, changes it or replaces it with a choice the record does not contain (Table T2). Kept: the cell, its operating point and logic vocabulary; $F = 2$ and buffer trees; the half adder and its “two half adders and an OR” full adder as reference; the record’s dissipation, noise-figure, noise-margin and classical-error laws, called read-only; and Part II’s Gaussian calculus (exact rules for loss, interference and phase-insensitive gain on coherent pulses) for every analog number. Five departures are substantive:

- **Slot rate.** 69 GHz gated, not Part I’s 0.25-THz baseline — the record’s verdict for $F = 2$ logic, not a choice (two-clock alternative: Section 8.1(e)).
- **Timing.** Part I’s DLY-cell balancing, generalized to a 92-cell-deep lane without latches or registers (Part I allows small fast registers), regenerates every waiting wire at every level (2,616 buffers, +51 %); stations every 10–13 cells, not 10–24; skew-tolerant alternatives open (Section 8.1(d)).
- **Decision error.** Part II gives one per-decision error (A0); Part I quotes none. This Part adds four compositions of the record’s pieces (A1–A3, B) and a yardstick, the model’s own Q8_0 error.
- **Analog unit.** Part II’s two-input QMAC-1, on a core Part II describes as four-quadrant (weights of either sign, Part II §4.1), becomes a three-input unsigned full-adder tile, QFA-3, with the sign in digital select logic, relaunched cell outputs [E7] as inputs, both readings of the unanchored rail plane, and $N_{op} = 2,600$ instead of 400.
- **Temperature.** The digital lane keeps the no-cryogenics lock if DG1 lands at A0 (Section 5.5); the analog lane is a Part-II cold-class design, priced with its cooler, as is a cooled digital lane.

Model-side alternatives — a lower activation precision, or a model taught to tolerate the fabric’s errors — leave the fabric unchanged and are priced in Section 8.2.

Construct	Parts I / II	Part III	Alternatives considered
Slot rate	0.25-THz design baseline (Part I); ≈ 0.1 THz flushed (note 07-23)	69 GHz [63, 74], the gated $F = 2$ frontier (note 07-31; neither confirmed nor moved at the bench’s ratio bias, notes 08-01, 08-02); 14.5 ps	two-clock menu: 0.25-THz transport over slower logic (§8.1(e))
Station spacing	“every 10–24 gates” (Part I §6.2)	10.0–13.3 cells by the released law; 24 unreachable (note 08-25)	none beyond 13 cells under the 20-dB link budget (§8.1(f))
Full adder	“two half adders and an OR” (Part I §6.5); 15 cells, depth 7 as composed here (§2.1)	kept as the reference idiom	threshold full adder, 5 cells, depth 2 (§8.1(a))
Path balancing	one DLY cell in the half adder	every waiting wire regenerated each level: 2,616 buffers	4- and 14-cell skew tolerance: 581, 189 (§8.1(d))
Fan-out	$F \leq 2$; buffer tiles for more (Part I §6.3)	kept; operand sharing by $F = 2$ trees [D4]	select in CMOS, nothing shared (§8.1(b))

Construct	Parts I / II	Part III	Alternatives considered
Memory	“compute in plasmons, store as charge”; small fast registers allowed (Part I §6.4)	no registers; nothing held on the fabric between slots (a departure: Part I allows small fast registers)	pumped bistable registers or recirculating loops for waiting carries (unmodelled, §8.1(d)); on-fabric coherent accumulation unbuildable (note 07-16)
Decision error	classical bit-error rate (BER) of a rail-restored decision (Part II)	booking A0, plus compositions A1–A3 and B [D5]	which applies is gate DG1 (§7)
Yardstick	none	the model’s own Q8_0 error, 42.5 dB (new)	error-aware fine-tuning to relax it (§8.2(b))
Analog unit	QMAC-1: 2 inputs, 2 comparators; sign asserted for the core (Part II §4.1)	QFA-3: 3 unsigned inputs, 3 comparators	phase-native trits; dual-rail ternary; $K \geq 4$ counters (§6.1–6.2)
Analog inputs	quiet comb launch of every input	relaunched cell outputs [E7] (new)	quiet launch as the PERM convention (§6.3)
Rail plane	rail anchored at no plane (note 07-17)	both carried: P2 (REF) and P1 (CONS) [E4] (§6.3)	—
N _{op}	400, “launch-budget-capped” (undefined)	2,600 at P2; 1,278 at P1	400 as a sensitivity: 2.3×10^{-2} at 4 K
Offset c	coded $1/\sqrt{12}$, open (note 07-15)	$1/\sqrt{12}$ in REF and CONS	$c = 0.02$ in PERM
Cell power	1.267 kW/cm ² stored at 353 K; kinetic 1.59–1.67 kW/cm ²	300 K by τ^{-3} scaling [D3]; 353 K kinetic	Joule model below 255 K, uncertified [E8]
Launch noise	20-dB link budget (Part I); 10 % coded jitter (Part II)	Part II’s jitter at the first decision after launch [D8]	proportional ($\times 4$) or rail-referenced ($\times 59$); $\times 239$ span (note 08-25)
Temperature	≤ 80 °C locked (Part I); cold classes as a fork (Part II)	digital lane warm; analog lane cold, with its cooler	cooled digital lane (§6.5)

Table T2. What Part III keeps, changes and considered instead, construct by construct. Bracketed labels [D...], [E...] name this paper’s in-model extensions (Table TA3); “§” without a Part refers to this paper.

3. The workload: Ternary Bonsai 2 27B

3.1 The model (external)

Everything in this subsection is external, from PrismML’s model card, configuration and reports [3–5] (Table T3). The model is derived from Qwen3.8-27B [10] with the architecture unchanged: 27.36 B parameters, 24.35 B in a 64-block language backbone, 2.54 B in the embedding table and LM head (the final projection onto the vocabulary) and 0.46 B in a vision tower (the card; its three parts sum to 27.35 B, and the whitepaper [4] gives 0.47 B). Three of every four blocks use Gated DeltaNet linear attention [16], a recurrent layer with a fixed-size state; every fourth uses full attention with a key-value (KV) cache. The hidden width is 5,120, the feed-forward network (FFN) width 17,408, the vocabulary 248,320.

The weights are ternary g128 (−1, 0 or +1, one 16-bit floating-point, FP16, scale per 128 weights), stored in a rotated basis: a fixed blockwise Hadamard rotation is folded into the weights and applied to the activations at runtime, as QuaRot [17] does against activation outliers. The representation costs 1.72 bits per weight: $\log_2 3 \approx 1.585$ bits per trit, plus the FP16 scale amortized over 128 weights, plus the few tensors kept in higher precision. In GGUF it ships in PrismML’s own packings, PTQ1_0 (dense trits at 1.75 bits per weight, 28 bytes per 128 weights; 5.95 GB) and PQ2_0 (2-bit slots, 7.21 GB); the whitepaper [4] quotes 1.76 bits per weight for the PTQ1_0 file. The model keeps 98.2 % of its FP16 parent’s average over 14 thinking-mode benchmarks, and decodes at 28.1 tokens per second (tok/s) on an Apple M5 Pro laptop (27.5-W GPU rail) and 129.9 tok/s on an RTX 5090 at 1.95 J per token. The whitepaper [4] and the release note [5] give slightly different throughput and quality figures (83.9 against 85.4 over a 20-benchmark suite; up to 143 tok/s on the RTX 5090; 46.8 tok/s on the M5 Max); none of them enters any computation here.

One assumption sets the lane’s precision. The card is silent on how its runtimes quantize activations. The lane adopts the Q8_0 block of the llama.cpp family [11] as its activation format — 32 8-bit integer (int8) values with one FP16 scale; whether PrismML’s kernels quantize per 32 or per 256 values (Q8_K, as mainline ggml does for its own ternary types [11]) is not stated (external, open). The lane’s cells are unchanged either way; only the host’s scale bookkeeping, and the yardstick of Section 5, would move. A ternary MAC is then $\text{int8} \times \{-1, 0, +1\}$, and a block of 32 products shares one activation scale: the unit mapped onto the fabric.

Item	Value	Source
Base model	Qwen3.8-27B, architecture unchanged	[3, 10]
Parameters	27.36 B = 24.35 B language backbone (64 blocks) + 2.54 B embedding/LM head + 0.46 B vision tower (27 blocks); the parts sum to 27.35 B, and [4] gives 0.47 B for the tower	[3]; [4]
Attention	hybrid, 3:1 — 48 Gated DeltaNet linear-attention blocks [16] and 16 full-attention blocks	[3], config
Widths	hidden 5,120; FFN 17,408 (SwiGLU); 24 query / 4 KV heads, head dimension 256	config
Linear-attention heads	16 key heads and 48 value heads of dimension 128 (q, k, v projection 2,048 + 2,048 + 6,144 = 10,240)	config
Vocabulary; context	248,320; 262 k tokens	[3], config
Weight format	ternary g128: $\{-1, 0, +1\}$, one FP16 scale per 128 weights; 1.72 bits per weight as a representation	[3]
Rotation	blockwise Hadamard (block 1,024, fixed ± 1 signs) folded into the weights; transform applied to activations at runtime [17]	[3]
Higher precision	26.2 M parameters (0.0976 % of the language model, not of the 27.36 B total): linear-attention recurrent-state path and normalization weights	[3]
Packings	GGUF files on disk: PTQ1_0 (dense trits, 1.75 bits per weight) 5.95 GB, PQ2_0 (2-bit slots) 7.21 GB [4] quotes 1.76 bits per weight and 5.93 GB for the PTQ1_0 file); MLX 2-bit 7.67 GB + 0.92 GB FP16 vision tower	[3]; [4]
Quality	98.2 % of FP16: 84.78 vs 86.32 over 14 thinking-mode benchmarks; math 96.57 vs 97.06; coding 89.42 vs 89.07	[3]
Decode, batch 1	≈ 47 tok/s Apple M5 Max; 28.1 tok/s M5 Pro (27.5-W GPU rail); 129.9 tok/s RTX 5090 at 1.95 J per token (PQ2_0); H100 SXM 86.9 / 113.9 tok/s (PTQ1_0 / PQ2_0)	[3]
Activations	not stated by the card; this paper assumes the llama.cpp family’s Q8_0 block (32 int8 + one FP16 scale); per-256 Q8_K, as mainline ggml pairs with its own ternary types, is possible (open); configuration dtype bfloat16	[11]; assumption

Table T3. Ternary Bonsai 2 27B, external facts from its model card, configuration and reports [3–5, 10]. Decode rates are the card’s token-generation measurements (tg128, batch 1); the Apple M5 Max figure is from the card’s earlier pre-rotation build. The whitepaper and release note differ slightly (Section 3.1); none of their figures enters a computation.

3.2 The per-token workload

From the external inputs [D6], digital_lane_bonsai.py §1 counts 25,598,361,600 ternary MACs per token (Table T4; shares computed from its rows): FFN blocks 67 %, Gated DeltaNet blocks 22 %, full attention 7 %, LM head 5 %. With the embedding table the model has 26,869,760,000 ternary weights, which checks exactly against the files (PQ2_0 spends 6 bytes more than PTQ1_0 per 128 weights). PTQ1_0 streams 5.669 GB per token (one embedding row is read, not the table). The Gated DeltaNet state (+314 MB, +5.5 %) and, at a 32 k context, the FP16 KV cache (+2.147 GB, +38 %) stay off the fabric. In the lane’s units a token is 799,948,800 block sums and 199,987,200 weight-group scales.

Matrix	Shape (out $\times$ in)	Count per token	Ternary MACs per token
GDN attn_qkv	10,240 $\times$ 5,120	48	2,516,582,400
GDN attn_gate	6,144 $\times$ 5,120	48	1,509,949,440
GDN ssm_out	5,120 $\times$ 6,144	48	1,509,949,440
Attention q + gate	12,288 $\times$ 5,120	16	1,006,632,960
Attention k	1,024 $\times$ 5,120	16	83,886,080
Attention v	1,024 $\times$ 5,120	16	83,886,080

Matrix	Shape (out × in)	Count per token	Ternary MACs per token
Attention o	5,120 × 6,144	16	503,316,480
FFN gate	17,408 × 5,120	64	5,704,253,440
FFN up	17,408 × 5,120	64	5,704,253,440
FFN down	5,120 × 17,408	64	5,704,253,440
LM head	248,320 × 5,120	1	1,271,398,400
Total			25,598,361,600
Ternary weights incl. embedding			26,869,760,000 (= (PQ2_0 – PTQ1_0 bytes)/6 × 128)
Weights streamed (PTQ1_0)			5.669 GB per token
GDN state read + write; FP16 KV at 32 k			+314 MB (+5.5 %); +2.147 GB (+38 %)
Block sums of 32; FP16 weight scales			799,948,800; 199,987,200

Table T4. The ternary workload of one decoded token (*digital_lane_bonsai.py* §1). GDN = Gated DeltaNet linear attention. Per-matrix counts multiply the listed shapes by the number of blocks; the total and the cross-check against the file sizes are printed by the listing.

3.3 The decisive property

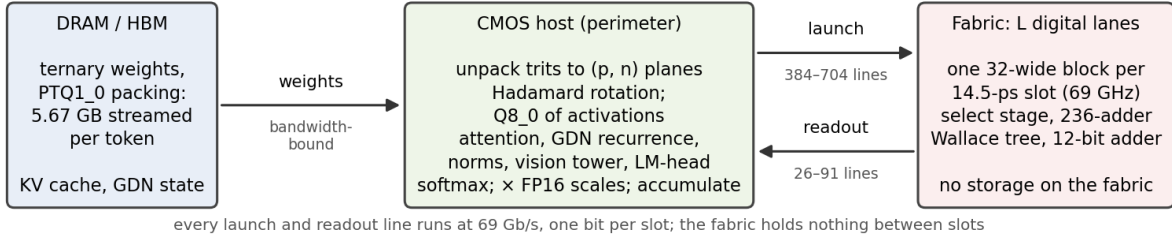
At batch 1, tokens per second are bandwidth over bytes per token [6, 7]: 88, 176 and 529 tok/s at 0.5, 1 and 3 TB/s (*digital_lane_bonsai.py* §5); the HBM3 high-bandwidth memory of an H100-class GPU delivers 3.35 TB/s [18] (external). No arithmetic unit can raise those rates; the fabric can only keep pace cheaply, in area and energy, and Section 4 is written to that specification.

The card’s measurements fit. The M5 Pro’s 28.1 tok/s from the 7.21-GB PQ2_0 file is $28.1 \times 7.21 \text{ GB} \approx 203 \text{ GB/s}$ if the whole file is read, as the card’s stated $\approx 204 \text{ GB/s}$ evidently assumes ($\approx 193 \text{ GB/s}$ without the embedding table, as in Section 3.2). On the H100, $86.9 \text{ tok/s} \times 5.67 \text{ GB} \approx 0.49 \text{ TB/s}$ is about 15 % of peak, which the card attributes to instruction throughput and launch overhead: unpacking trits costs arithmetic, which is why the host does it here (Section 4.1). *digital_lane_bonsai.py* §5 also prints three reported rates, RTX 4090 86.0 tok/s (PTQ1_0; the card gives 91.1) and Section 3.1’s press figures, 46.8 and 143 tok/s; none enters a computation.

4. The embedding: host, lanes and the digital lane

4.1 System partition

Figure T1 shows the partition. Weights stream from DRAM or HBM in the PTQ1_0 packing to a CMOS host at the fabric’s perimeter, which unpacks each trit into two bit planes, $p = [w = +1]$ and $n = [w = -1]$; applies the Hadamard transform (21.7–29.1 M additions per token) and Q8_0 quantization to the activations; and keeps every stateful or floating-point task — attention against the KV cache (0.81 G MACs per token at a 4 k context), the Gated DeltaNet recurrence ($\approx 151 \text{ M FP32 MACs}$), norms, the vision tower and the LM-head softmax (*digital_lane_bonsai.py* §5). Each slot, a lane takes one 32-value activation block and the 32 matching weights of a row, and returns a 13-bit unsigned block sum U, which the host converts, scales and accumulates. Nothing is held on the fabric between slots.



What sets the speed (in-model, Section 4.6)

- tokens per second = memory bandwidth ÷ 5.67 GB: 88, 176, 529 tok/s at 0.5, 1, 3 TB/s
- one lane = 32 MACs per slot at 69 GHz = 86 tok/s of the 25.6 G MACs per token; 2, 3, 7 lanes keep pace
- the fabric adds area and energy, not speed: 17 k to 62 k cells, about 1.7 mm² at most, 0.13 to 0.64 W at 300 K
- the host scales 800 M block sums per token (1.27 TFLOP/s at 529 tok/s) and launches 27 to 49 Tb/s of operands

Figure T1. The system partition (Section 4.1). DRAM or HBM streams the PTQ1_0-packed ternary weights (5.67 GB per token; the KV cache and the Gated DeltaNet state also live there) over a bandwidth-bound link to a CMOS host at the fabric's perimeter. The host unpacks trits into (p, n) bit planes, applies the Hadamard rotation and Q8_0 quantization to the activations, runs attention, the GDN recurrence, norms, the vision tower and the LM-head softmax, and applies the FP16 group scales and accumulates. The fabric holds L digital lanes, each computing one 32-wide block per 14.5-ps slot (69 GHz) — select (p AND a) OR (n AND NOT a), a Wallace tree of 236 full adders and a 12-bit adder to a 13-bit block sum — with no storage on the fabric; 384–704 launch lines and 26–91 readout lines run at 69 Gb/s, one bit per slot. The text below the diagram states what sets the speed (Section 4.6): memory bandwidth ÷ 5.67 GB gives 88, 176 and 529 tok/s at 0.5, 1 and 3 TB/s; one lane is 32 MACs per slot at 69 GHz, 86 tok/s of the 25.6 G MACs per token, so 2, 3 and 7 lanes keep pace; the fabric adds area and energy, not speed (17–62 k cells, at most about 1.7 mm², 0.13–0.64 W at 300 K at the hydrodynamic tier); the host scales 800 M block sums per token (1.27 TFLOP/s at 529 tok/s) and launches 27–49 Tb/s of operands. In-model; nothing shown has been built.

No multiplier is needed because of a select identity. Each product bit is $x_m = (p \wedge a_m) \vee (n \wedge \neg a_m)$, $m = 0 \dots 7$: the product is a for $w = +1$, 0 for $w = 0$, and the one's complement $\neg a = -a - 1$ for $w = -1$, the missing +1 being folded into a per-block constant n_neg , the number of -1 weights in the block. Inverting each product's sign bit adds 128 and makes it an unsigned $u \in [0, 255]$, so the block sum $U = \sum u$ satisfies $0 \leq U \leq 8,160 < 2^{13}$, and the host recovers the dot product with two constants:

$$u_i = [(p_i \wedge a_i) \vee (n_i \wedge \neg a_i)] \oplus 128, \quad 0 \leq U = \sum_i u_i \leq 8,160, \quad \sum_i w_i a_i = U + n_neg - 32 \cdot 128 \quad (T1)$$

The rest of the lane only adds bits, with the full adder, or 3:2 counter, which turns three bits of equal weight (inputs i_1, i_2 and a carry in c_in) into a sum bit s and a carry bit c_out of double weight while preserving their count:

$$s + 2 \cdot c_out = i_1 + i_2 + c_in \quad (T2)$$

4.2 The lane netlist

The lane [D1] computes one block sum per slot (digital_lane_bonsai.py §2; Table T5; Figure T2). Its select stage builds $(p \wedge a) \vee (n \wedge \neg a)$ from four cells per product bit — two ANDs, a NOT and an OR — which is 1,024 cells for 32 products of 8 bits, and 32 NOT cells invert the sign bits. Each weight wire must reach eight select cells, which at $F = 2$ takes a six-cell buffer tree: 384 cells for the 64 wires. The 256 product bits then stand in eight columns of 32, one per bit weight.

A Wallace tree [14, 15] reduces the columns: each stage takes every column's bits three at a time into full adders, keeping sums in the column and moving carries one column up, until no column holds more than two bits. Eight columns of 32 take 236 full adders in 8 stages (per bit column: {0: 15, 1: 23, 2: 27, 3: 29, 4–7: 30 each, 8: 14, 9: 6, 10: 2}), and a 12-bit ripple adder yields the 13-bit result. At 15 cells per full adder the logic is $1,024 + 32 + 384 + 248 \times 15 = 5,160$ cells, all Part-I cells used as Part I uses them, with fan-out at most two.

Two features are choices: the block width, Q8_0's 32 (Section 8.1(c)), and the record's composed full adder rather than a cheaper threshold one the record does not contain (Section 8.1(a)).

Element	Cells	Note
Select stage	1,024	4 cells per product bit: $(p \wedge a) \vee (n \wedge \neg a)$, 32×8 bits
Sign-bit NOT	32	makes every product unsigned, eq. (T1)
Weight fan-out trees	384	64 wires $\times$ 6 cells, $F = 2$
Wallace tree	3,540	236 full adders $\times$ 15, 8 stages, 87 pass-through bit-stages
Ripple adder	180	12 full adders $\times$ 15; 13-bit output
Logic	5,160	record idiom (2 half adders + OR = 15 cells per full adder)
Balancing buffers, tolerance 0 / 4 / 14 cells	2,616 / 581 / 189	+51 % / +11 % / +4 % on logic
Re-sync stations, spacing 13.3 / 10.0 cells	393 / 679	one per wire crossing each cut
Timing-closed lane, tolerance 0 (reference)	8,169–8,455	0.224–0.232 mm ² at 27.4 μ m ² per cell
Timing-closed lane, tolerance 4 / 14 cells	6,134–6,420 / 5,742–6,028	if the gated cell tolerates the skew (§8.1(d))
Depth model	5,472 wires	critical path 92 cells ($\approx$ 31 ps at $\approx$ 0.33 ps per cell, Part I's 4-ps baseline; 2.1 slots); longest wait 44 cells; outputs at depths 26–92

Table T5. The lane's cell budget (*digital_lane_bonsai.py* §2; in-model). The lane range spans the two station spacings, 13.3 cells (noise-figure floor) and 10.0 cells (NF = 4 dB).

4.3 Timing closure

On a wave-pipelined fabric every signal must reach its consumer in the slot in which it is needed. The depth model [D7] builds the netlist's 5,472 wires with one cell delay per gate (*digital_lane_bonsai.py* §2). The critical path is 92 cells, $\approx$ 31 ps or 2.1 slots at the $\approx$ 0.33 ps per cell of Part I's half adder at its 4-ps baseline ($\approx$ 1 ps at depth 3, Part I Table 5). At the 69-GHz gated point the per-cell latency is not established — a cell's transient gain is set by drive duration (Part I §6.2), here a duty-0.8 gated burst (note 2026-07-31) — so the number of blocks in flight is open (gate DG2); the cell counts do not depend on it. The 13 output bits leave at depths 26–92, each on its own readout line, and are deskewed at the receiver.

A carry produced early may wait up to 44 cells for its consumer. This lane has no registers (Section 2.1), a re-sync station re-times a pulse but cannot hold one, and a plasmon decays in about a picosecond, so the bit is regenerated at every level it waits, one buffer per cell delay, exactly as Part I's DLY cell does. At zero skew tolerance this costs 2,616 balancing buffers, +51 % on the logic; a 4- or 14-cell tolerance would need 581 or 189, but the record is silent on a gated cell's delay spread, so the reference takes zero (Section 8.1(d)).

Stations are counted as one per wire crossing each cut of the depth axis: 679 at 10.0-cell spacing and 393 at 13.3 (the earlier rule of one station per 10–24 logic cells gave 215–516). The timing-closed reference lane is $5,160 + 2,616 + (393 \text{ to } 679) = 8,169\text{--}8,455$ cells, 0.224–0.232 mm² at 27.4 μ m² per cell (Table T5; Figure T2B). Its critical path crosses 7–9 stations, which sets the length of gate DG2 (Section 7).

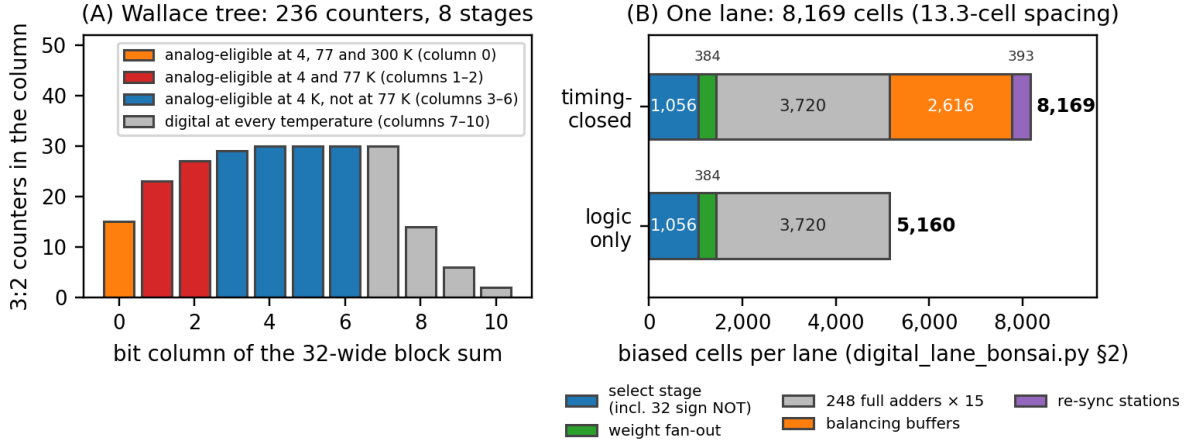


Figure T2. The lane (Sections 4.2–4.3 and 6.4). (A) The Wallace tree: 3:2 counters per bit column of the 32-wide block sum, 236 counters in 8 stages. Colours mark which columns’ counters could be analog QFA-3 tiles at 41-dB dot-product SNR under the reference conventions (*ternary_qmac_design.py* §5, REF, Student-t activations), in three non-overlapping groups set by b^* , the highest bit column whose counters can be analog: column 0 at 4, 77 and 300 K ($b^* = 0$ at 300 K); columns 1–2 at 4 and 77 K ($b^* = 2$ at 77 K); columns 3–6 at 4 K but not at 77 K ($b^* = 6$ at 4 K; 5 at 20 K and 3 at 48 K, Table T8); columns 7–10 are digital at every temperature. (B) One lane’s cell budget (*digital_lane_bonsai.py* §2): logic only, 5,160 cells, against the timing-closed lane at the 13.3-cell station spacing, every segment labelled — select stage including the 32 sign-bit inverters (1,056), weight fan-out (384), 248 full adders (236 tree + 12 ripple) $\times 15 = 3,720$, balancing buffers (2,616) and re-sync stations (393), 8,169 cells in all; at the 10.0-cell spacing the lane is 8,455 cells (0.224–0.232 mm² across the two). In-model.

4.4 Bit-exactness

The lane is exact by construction: every full adder preserves its count (T2), every product enters the tree as an unsigned u (T1), and $U \leq 8,160 < 2^{13}$ cannot overflow 13 bits.

Demonstrated (runnable); a netlist check, not physics. *digital_lane_bonsai.py* §3 runs Bonsai-shaped matrices through the gate-level netlist with Boolean decisions — the full-attention k projection (163,840 block sums, 5.2 M MACs), the full Gated DeltaNet *ssm_out* projection (983,040; 31.5 M) and 1,024 rows of an FFN down projection (557,056; 17.8 M) — and every block sum is exact, the host’s scaling reproducing the int8 reference bit for bit. The weights are synthetic and ternary, with an assumed one-third share of zeros; the activations are Gaussian and Q8_0-quantized, so the only gap to unquantized floating point is Q8_0’s own 45.0–45.6 dB. This checks the netlist and the host’s arithmetic; whether real cells decide correctly often enough is Section 5’s question.

4.5 Power, energy, heat

At 300 K, with the hydrodynamic-tier τ^{-3} -scaled dissipation [D3] and no kinetic correction to it (the record does not evaluate that correction at 300 K; at 353 K it raises the dissipation $\times 1.25$ –1.32, Part I §8.2), a lane draws 64–86 mW, 29–39 fJ per MAC at full utilization; at the 353 K bias, kinetic-corrected, it draws 130–141 mW, 59–64 fJ per MAC (*digital_lane_bonsai.py* §2; in-model). The 69-GHz slot was adjudicated at 353 K and is unmodelled at 300 K. Heat is no constraint: at a 3.6 % fill the lane rises 2.8–3.7 K (300 K) or 5.8–6.1 K (353 K) against Part I’s 55 K budget (*digital_lane_bonsai.py* §5); the configurations of Section 4.6 spend 0.9–3.3 mJ per token.

That is not where a decode’s energy goes: reading 5.67 GB from HBM at ≈ 4 pJ per bit (HBM2-class, [19]) costs 5.67 GB $\times$ 8 bit/B $\times$ 4 pJ/bit $\approx$ 0.18 J per token, so the fabric adds about 0.5–2 %, and a GPU decode measures 1.95 J per token of board power (RTX 5090; external). Not booked are the CMOS perimeter and the wall-plug launch energy, which the record cannot fix because the launch-coupling efficiency is unquantified (Part I §9.5; the gated burst costs $\times 3.4$ –4.9 energy per ‘1’ over the released burst,

note 2026-07-31). The claim, if the cell works, is that the fabric’s share of the arithmetic energy is small — not that the system is efficient.

4.6 The system

One lane does 32 MACs per slot, 2.208×10^{12} MACs per second at 69 GHz, or 86.26 tok/s of Bonsai’s 25.6 G MACs per token (78.8–92.5 tok/s over 63–74 GHz). The memory sets the token rate, so the lane count follows from bandwidth (Table T6; Figure T5A; `digital_lane_bonsai.py` §5): 2 lanes at 0.5 TB/s (51 % busy), 3 at 1 TB/s (68 %), 7 at 3 TB/s (88 %). Lanes compute different rows against the same activation block, which is launched once and shared through $F = 2$ buffer trees of $256 \cdot (2L - 2)$ cells [D4]. Seven lanes take 60,255–62,257 cells, 1.65–1.71 mm² and 0.47–0.64 W at 300 K (0.96–1.04 W at 353 K).

The input–output count is the harder number: 64 weight and 13 readout lines per lane plus 256 shared activation lines — at 3 TB/s, 704 launch lines (48.6 Tb/s) and 91 readout lines (6.28 Tb/s) at 69 Gb/s each, 3.14 launched bits per MAC. The ≈ 800 lines — modulated, phase-locked and coupled into the acoustic plasmon at an efficiency the record does not quantify — are harder engineering than the lane.

The host’s load is modest: scaling 800 M block sums per token at three operations each is 2.40 GFLOP (1.27 TFLOP/s at 529 tok/s), and the `n_neg` counts, fixed by the weights, can be precomputed and streamed (+10.6 % traffic) or counted during unpacking. Within 10^5 cells — the top of Part I’s illustrative 10^4 – 10^5 -cell scale (Part I §8.1), not a cap — 11 lanes give 949 tok/s, which a 3-TB/s memory feeds only with a batch of two or more.

Bandwidth	tok/s	Lanes (busy)	Cells; area	Fabric W at 300 K / 353 K	mJ per token	Launch lines (Tb/s)	Readout lines (Tb/s)	Launched bits per MAC
0.5 TB/s	88.2	2 (51 %)	16,850–17,422; 0.46–0.48 mm ²	0.13–0.18 / 0.27–0.29	1.5–3.3	384 (26.5)	26 (1.79)	6.00
1.0 TB/s	176.4	3 (68 %)	25,531–26,389; 0.70–0.72 mm ²	0.20–0.27 / 0.41–0.44	1.1–2.5	448 (30.9)	39 (2.69)	4.67
3.0 TB/s	529.2	7 (88 %)	60,255–62,257; 1.65–1.71 mm ²	0.47–0.64 / 0.96–1.04	0.9–2.0	704 (48.6)	91 (6.28)	3.14

Table T6. The system at three weight-streaming bandwidths (`digital_lane_bonsai.py` §5; in-model plus arithmetic). Tokens per second are memory-bound (5.669 GB per token); one lane is 86.26 tok/s at 69 GHz. Cells include the $F = 2$ activation-sharing trees. Energy per token spans 300 K (low) to 353 K kinetic-corrected (high). Every line runs at 69 Gb/s.

5. The error budget: what “working” means

Section 4 shows that the lane is right if its cells decide correctly. This section asks how often they may not, against a yardstick taken from the model itself (`digital_lane_bonsai.py` §4; Table T7; Figure T3).

5.1 The yardstick

This paper takes the error the runtime already accepts: Q8_0 quantization of the activations (the format assumed in Section 3.1) perturbs every dot product, so a lane adding no more error power at most doubles that perturbation. (The card’s scores were measured with vLLM, so what Q8_0 itself costs in quality is not reported.) Over 4,000 synthetic dot products of length $K = 5,120$, Q8_0’s signal-to-error ratio is 45.5 dB for Gaussian and 42.5 dB for Student-t ($v = 4$) activations; the reference uses 42.5 dB. This is a proxy: Bonsai’s tolerance of lane errors is open, perhaps larger (it was trained against aggressive quantization) or smaller (lane errors are rare and large, not dense and small).

5.2 Error propagation and the pass line

A wrong decision in a high column changes the block sum by up to 2^{12} , one in column 0 by 1. `digital_lane_bonsai.py` §4 measures the average by gate-level emulation [D2], flipping every cell decision

independently with probability BER (the error rate per decision). Over 163,840 blocks at $BER = 1 \times 10^{-5}$, 2×10^{-5} and 4×10^{-5} the block-sum error variance per unit BER is $\kappa = 2.8 \times 10^8 \text{ int}^2$ (13 % spread); flipping launched bits instead gives $6.7\text{--}6.9 \times 10^5$. A $K = 5,120$ dot product is $N_B = 160$ blocks, so the lane's error power is $N_B \cdot \kappa \cdot BER$, and equating it to $Q8_0$'s gives the pass line:

$$BER_{ceil} = P_{sig} / (N_B \cdot \kappa \cdot 10^{(SNR_q/10)}) \quad (T3)$$

where P_{sig} is the dot-product signal power in the lane's integer units and $SNR_q = 42.5 \text{ dB}$: 6.3×10^{-9} per cell decision and 2.6×10^{-6} per launched bit (in-model; emulation runnable), or 6.3×10^{-10} per decision for a 10-dB margin below $Q8_0$. The ceiling is an upper bound. The emulation flips only the $\approx 5,100$ logic decisions of the netlist (8,361 flips in 163,840 blocks at 10^{-5}); the timing-closed lane adds 2,616 balancing buffers and 393–679 stations, which also decide and carry the weight of the wire they regenerate. Its κ is larger by an uncomputed factor, so 6.3×10^{-9} per decision, and every pass temperature below, is an upper bound (Section 9, item 6).

5.3 Five bookings of one decision

The record gives one error rate and implies several. Five bookings are carried, all assembled from released pieces [D5] (Table T7; Figure T3):

- **A0, released.** Part II's classical error of a rail-restored decision (qerrors.classical_ber: $k = 8$, full rail swing, matched noise of one restoring cell): 3.6×10^{-11} at 300 K (lane SNR 64.9 dB) and 9.1×10^{-10} at 353 K (50.9 dB). It passes below 394 K.
- **A1, F = 2 softening.** The same law at $k_{eff} = 8/\sqrt{2}$, the softening note 2026-07-21 derives for a cell driven through an $F = 2$ split (as the lane's fan-out trees, select stage and half adders are), applied to every decision: 9.7×10^{-9} (40.6 dB) and 1.1×10^{-7} (30.1 dB). It passes below 292 K.
- **A2, split.** A1 decided on the swing a driven cell receives after a -3 dB split: 3.6×10^{-5} (5.0 dB) and 1.2×10^{-4} (-0.4 dB). It passes below 144 K.
- **A3, junction.** A2 through one junction at the -1 dB optimistic end of Part I's own worst path, "launch $\rightarrow -3 \text{ dB}$ split $\rightarrow +8 \text{ dB}$ cell $\rightarrow -1\text{...}-3 \text{ dB}$ junction" (Part I Table 5): 2.0×10^{-4} (-2.5 dB) and 5.4×10^{-4} (-6.9 dB). It passes below 114 K; the -3 dB end is not computed and passes only colder.
- **B, Part II's calculus.** The Gaussian calculus with relaunched inputs [E4, E7], applied to a half adder read as a two-input interference decoded at the rail: 5.5×10^{-3} per decode at 300 K (-4.0 dB by emulation), 1.0×10^{-2} at 353 K (-6.7 dB by emulation, -6.6 dB mapped) and 1.2×10^{-6} at 77 K (32.6 dB). It passes only below 59 K (digital_lane_bonsai.py §4: reference lane, 42.5-dB pass line). The 77 K SNR is mapped onto the per-decision axis through the 300 K emulation (0.051 equivalent cell errors per decode error), since the direct 77 K emulation prints 35.4 dB from only ≈ 25 error events; the analog listing's 32.4 dB (Section 6.6) agrees with the mapped value within 0.3 dB.

At 77 K every A booking is far inside the line (5.7×10^{-37} to 3.2×10^{-12}), while B, at 32.6 dB, is still outside. None of A1–A3 or B is a record value; each composes record pieces as the record does not, which is why they are bookings, not predictions.

Booking	Assumes	300 K: error (SNR)	353 K: error (SNR)	77 K: error (SNR)	Passes below
A0	released classical BER: $k = 8$, full rail swing, matched noise of one restoring cell (Part II)	3.6×10^{-11} (64.9 dB)	9.1×10^{-10} (50.9 dB)	5.7×10^{-37} (322.9 dB)	394 K
A1	A0 at $k_{\text{eff}} = 8/\sqrt{2}$, the $F = 2$ softening (note 07-21)	9.7×10^{-9} (40.6 dB)	1.1×10^{-7} (30.1 dB)	5.6×10^{-28} (233.0 dB)	292 K
A2	A1 on the swing after a -3 dB split	3.6×10^{-5} (5.0 dB)	1.2×10^{-4} (-0.4 dB)	6.3×10^{-15} (102.5 dB)	144 K
A3	A2 through one -1 dB junction (Part I's worst path)	2.0×10^{-4} (-2.5 dB)	5.4×10^{-4} (-6.9 dB)	3.2×10^{-12} (75.5 dB)	114 K
B	Part-II calculus [E4, E7] on a rail-referenced half-adder decode (per decode)	5.5×10^{-3} (-4.0 dB)	1.0×10^{-2} (-6.7 dB)	1.2×10^{-6} (32.6 dB, mapped)	59 K
Pass line	lane error power = Q8_0's (Student-t); 10-dB margin: 6.3×10^{-10}	6.3×10^{-9} (42.5 dB)	6.3×10^{-9} (42.5 dB)	6.3×10^{-9} (42.5 dB)	—

Table T7. Five bookings of one cell decision (*digital_lane_bonsai.py* §4; in-model). Errors are per cell decision (per half-adder decode for B); the bracketed SNR is the lane's digital-error SNR for a $K = 5,120$ dot product with Student-t activations. Pass temperatures are the highest at which the booking meets the pass line (upper bounds for the timing-closed lane; Section 5.2). Row B's SNRs at 300 and 353 K are the direct emulation; mapped onto the per-decision axis through the 300 K emulation (0.051 equivalent cell errors per decode error, $KAPPA_B_RATIO$), they read -4.0 dB (identical by construction) and -6.6 dB. At 77 K, where the direct emulation sees only ≈ 25 error events and prints 35.4 dB, the table gives the mapped 32.6 dB, within 0.3 dB of the analog listing's 32.4 dB (Section 6.6).

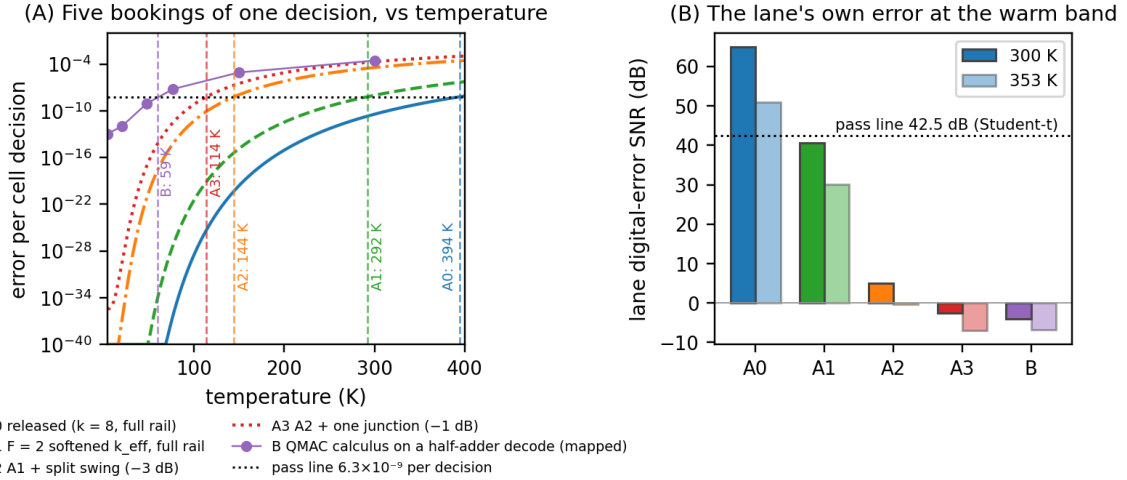


Figure T3. The error budget (Section 5; *digital_lane_bonsai.py* §4). (A) Error per cell decision versus temperature for five bookings of one decision, told apart by line style: A0 (released, $k = 8$, full rail; solid), A1 ($F = 2$ -softened k_{eff} , full rail; dashed), A2 (A1 decided on the -3 dB split swing; dash-dot) and A3 (A2 plus one -1 dB junction; dotted), computed from the released chains; and B (the Part-II calculus applied to a half-adder decode; circles on a thin line) at 4, 20, 48, 77, 150 and 300 K, the per-decode errors of *ternary_qmac_design.py* §6 mapped onto the per-decision axis by the 300 K emulation's factor, 0.051 equivalent cell errors per decode error ($KAPPA_B_RATIO$, *digital_lane_bonsai.py* §4). The black dotted line is the pass line, 6.3×10^{-9} per decision (lane error power equal to Q8_0's, Student-t activations); the labelled dashed verticals mark the pass temperatures, A0 394 K, A1 292 K, A2 144 K, A3 114 K and B 59 K; the legend sits below the axes. (B) The lane's digital-error SNR at 300 K (solid) and 353 K (pale) for the five bookings against the pass line, 42.5 dB (Student-t); only A0 clears it. In-model.

5.4 Launch noise

Part II's coded launch — 20 dB, read as amplitude jitter of 10 % of the swing — added at the first decision after each launch [D8] makes each launched bit err 8.5×10^{-3} ($k = 8$) to 2.0×10^{-2} (k_{eff}) at 300 K,

more than three orders of magnitude above the 2.6×10^{-6} ceiling. Meeting it needs a launch SNR of 27.9 dB ($k = 8$) or 30.9 dB (k_{eff}) at 300 K and 28.7 or 32.7 dB at 353 K — jitter at most 4.0, 2.8, 3.7 or 2.3 % of the swing (digital_lane_bonsai.py §4) — on all 384–704 lines: Part II’s quiet-launch rule holds for a digital lane too. Note 2026-08-25 shows the 20-dB figure means different quantities in Parts I and II; its launcher-physics fork (the note’s check P3), run on Part II’s decode, spans $\times 239$ in decode error across four launcher conventions — at the coded N_{op} plane a proportional jitter lowers the error about fourfold, and a rail reference raises it up to 59-fold (open until the launcher’s physics and reference plane are known). The figures above use Part II’s additive convention at the N_{op} plane.

5.5 Reading the budget

Three readings follow, all in-model. Only the released booking A0 passes warm, with 22 dB of margin at 300 K and 8 dB at 353 K. The record’s own $F = 2$ softening, hard to avoid in a lane of $F = 2$ trees and half adders, puts the lane at the Q8_0 level: 40.6 against 42.5 dB at 300 K, a 1.9-dB fail, passing below 292 K. And if a decision’s swing is what the driven cell receives after Part I’s worst-path split and junction — the budget Part I uses for gain — the pass temperature falls to 144 K for the split alone and to 114 K through a -1 dB junction (colder at -3 dB, not computed). Which reading holds is the open question of which reference plane a decision’s swing belongs to (notes 2026-07-15, 2026-07-17).

The lane, exact as a netlist, therefore works warm only if a loaded cell decides as well as the released booking says — one measurement, a half adder’s decision error at fan-out 2, through real junctions, at rail drive, at 300 K and 353 K: gate DG1 (Section 7). At A0 the lane is a room-temperature design; at A1 it wants about 8 K of cooling below 300 K or about 2 dB more model tolerance (61 K or 12.4 dB at the 353 K bias; Section 8.2); at A2–A3 it is a 114–144 K design, reopening Section 6.5; at B it is cryogenic.

6. The analog alternative: the Part-II fabric as a counter tree

Part II’s fabric adds by interference — pulses superpose passively at a junction and threshold cells decode the sum — so interference might replace some of the Wallace tree’s 236 full adders; ternary_qmac_design.py asks whether (Tables T8, T9; Figure T4). Every number here carries one caveat: a second independent check of that listing raised five rework items that remain open (Appendix TA).

6.1 Why the phase-native trit is tempting, and why it loses

The tempting design puts the weight into the physics: $+1$ launches the activation’s pulse at phase 0, -1 at phase π , 0 not at all, and interference forms the signed sum. Part II asserts sign-programmable weights only in text (Part II §4.1); no released call computes a signed tile. A signed tile built here [E6] with a coherent offset arm, summing two signed trits (five levels on four comparators), errs 5.4×10^{-4} at 4 K and 3.3×10^{-2} at 77 K, against 3.5×10^{-7} and 1.3×10^{-3} for the unsigned three-input counter below under the same quiet conventions (ternary_qmac_design.py §3): a signed sum crowds its levels around the offset, which also spends rail headroom.

Dual-rail balanced ternary — separate rails for $+1$ and -1 contributions, recombined by powers of three — fares worse: five trits span only ± 121 , so int8 needs a sixth trit or a requantization, and even lower-bound counts are 387–509 counters per block against 236 for binary columns (1.64 – $2.16\times$). The sign therefore stays arithmetic, in the select stage, and every analog input is an unsigned bit — a departure from the four-quadrant core Part II asserts for QMAC-1.

6.2 The K-input counter and the split law

An unsigned K-input counter interferes K pulses of N_{op} quanta in a balanced combiner tree [E1] and must separate $K + 1$ levels with K comparators. Each comparator needs the whole sum, so a K-way fan-

out divides it (extension [E2] charges 3 dB per halving and $\lceil \log_2 K \rceil$ junction levels), and the per-comparator level spacing obeys a split law (ternary_qmac_design.py §2):

$$\Delta^2(K) = (2N_{op}/K) \cdot \eta_{in} \cdot \eta_{extra} \cdot \eta_{fo}(K) \rightarrow 2N_{rail} \cdot \eta_{fo}(K)/K^2 \propto K^{-3} \quad (T4)$$

where Δ is the level spacing in Part II's quadrature units (field amplitude, vacuum variance $\frac{1}{2}$), $N_{rail} \approx 3,833$ quanta is the rail, and η_{in} , η_{extra} , η_{fo} are the transmissions of the input transit, extra junction levels and comparator fan-out. The arrow caps the top level at the rail at the combiner output (plane P2), and the K^{-3} is rough, since η_{fo} carries the $1/K$ split; at P2, Δ^2 at 77 K with a quiet launch falls as 673, 140, 59 and 5.2 for $K = 2, 3, 4$ and 8. A counter must also remove bits — K in, $\lceil \log_2(K + 1) \rceil$ out — so $K = 2$ (QMAC-1, a half adder) removes none and $K = 3$ or 4 removes one. At $K = 2$ the tile reproduces Part II's qmac.error_2bit bit for bit at 48 points spanning temperature, N_{op} , junction loss, launch noise and propagation convention (demonstrated (runnable); ternary_qmac_design.py §1). Under REF (below), $K = 3$ errs 1.3×10^{-5} , 3.9×10^{-5} , 5.0×10^{-3} and 0.15 at 4, 20, 77 and 300 K, while $K = 4$ already errs 1.8×10^{-3} at 4 K and $K \geq 4$ stays above 10^{-6} with every optimistic lever (2.4×10^{-6} under PERM). The tile to build is $K = 3$.

QFA-3 is that tile: three pulses interfere passively, a 1:3 split feeds three comparators, and the thermometer code $t_1 t_2 t_3$ ($t_i = 1$ if at least i inputs are 1) becomes $\text{sum} = (t_1 \wedge \neg t_2) \vee t_3$ and $\text{carry} = t_2$. In the record's idiom that is three comparators, a NOT, an AND, an OR and three path-balancing buffers — 9 biased cells — plus three optional carry-alignment buffers, 12 in all, which the listing charges against the full adder's 15. The comparators inherit Part II's $k \geq 16$ redesign, not charged here.

6.3 Conventions and the operating envelope

Three conventions are carried, because the record leaves three things open (ternary_qmac_design.py §2, §4):

- **REF.** The rail caps the top level at the combiner output (P2, the first plane where the sum is one field). Inputs are relaunched [E7]: inside a tree each input is an earlier cell's output, so a '1' is a rail pulse attenuated to N_{op} and a '0' a cell's unsaturated output carrying the released amplifier's noise, $\sigma_0 = 3.56\%$ of the rail swing at 4 K. Extra junction levels add a quarter-wave each; $c = 1/\sqrt{12}$ enters the decision's static-offset term [E3].
- **CONS.** REF with the rail capping the pre-loss launch sum (plane P1), so $N_{op} = N_{rail}/3 = 1,278$.
- **PERM.** A quiet comb launch, P2, multiport junctions, junction-only extra levels and $c = 0.02$.

Under REF, $N_{op} = 2,600$ quanta from 4 to 150 K — 6.5 times the registered 400 — set by a 1.69-dB relaunch attenuator from the rail, at 1.72 aJ per pulse (Table T8). The error stays $\leq 10^{-3}$ below 49.6 K (REF), 20.5 K (CONS) or 126.5 K (PERM), and $\leq 10^{-4}$ below 27.2 K, never, or 88.1 K; 10^{-5} and 10^{-6} are reached only under PERM, below 66.4 and 52.3 K.

From REF's 1.3×10^{-5} at 4 K, one change at a time: a quiet launch gives 3.5×10^{-7} , $c = 0.02$ gives 8.4×10^{-7} , multiport junctions 2.2×10^{-6} , the P1 plane 4.0×10^{-4} , the registered N_{op} (400) 2.3×10^{-2} , and -3 dB junctions 9.9×10^{-4} (flagged by the second check). The rail plane alone moves the error 30-fold, and the record anchors it nowhere. A tile at 4 K stays within $2\times$ of its quiet-launch error (3.5×10^{-7}) only if a '0' level's noise is $\sigma_0 \leq 1.56\%$ of the rail swing, where the released amplifier rule gives 3.56 %, and the launch jitter is $\sigma_{rel} \leq 1.33\%$ of the launched amplitude ($S \geq 37.5$ dB); at the coded 20 dB the error is pinned near 1.1×10^{-2} .

T (K)	REF: N _{op} ; p	CONS: p	PERM: p	All-QFA tree SNR (REF)	b* (REF)	Analog counters REF / CONS / PERM
4	2,600; 1.3×10 ⁻⁵	4.0×10 ⁻⁴	4.1×10 ⁻¹³	28.2 dB	6	184 / 94 / 236
20	2,600; 3.9×10 ⁻⁵	9.6×10 ⁻⁴	2.5×10 ⁻¹¹	23.4 dB	5	154 / 94 / 236
48	2,600; 8.8×10 ⁻⁴	9.3×10 ⁻³	4.1×10 ⁻⁷	9.9 dB	3	94 / 38 / 236
77	2,600; 5.0×10 ⁻³	3.0×10 ⁻²	3.5×10 ⁻⁵	2.4 dB	2	65 / 38 / 154
150	2,600; 3.3×10 ⁻²	0.10	2.4×10 ⁻³	-5.8 dB	1	38 / 15 / 65
300	3,338; 0.15	0.32	4.1×10 ⁻²	-12.3 dB	0	15 / 0 / 15

Table T8. The QFA-3 tile and the hybrid lane (ternary_qmac_design.py §4–§5; in-model). p is the QFA-3 symbol error (not the weight bit plane p of Section 4); N_{op} is 1,278 throughout under CONS and 1,822 (2,065 at 300 K) under PERM. The SNR is that of a 5,120-long dot product ($N_B = 160$ blocks) with every counter analog (Student- t activations); b^* is the highest bit column whose counters can be analog while the dot product keeps 41 dB, and the last column counts those counters out of 236.

6.4 The hybrid lane

The hybrid lane keeps the 236-counter tree of Section 4.2, makes the counters of columns 0 ... b^* QFA-3 tiles and leaves higher columns digital, since a wrong decode in column j costs 2^j . The functional tree is exact on 20,000 random blocks (demonstrated (runnable); a netlist check, not physics), and the error model matches Monte Carlo to 1.6 % (ternary_qmac_design.py §5). The analog listing uses its own Q8_0 estimator (42.7–44.6 dB) and a 41-dB target, 1.5 dB below Section 5's 42.5-dB pass line; the counts below are at that target, and an open rework item finds its Student- t SNRs 1.7 dB optimistic (0.2 dB for Gaussian; Appendix TA).

An all-QFA tree reaches only 28.2, 23.4, 9.9, 2.4, -5.8 and -12.3 dB at 4, 20, 48, 77, 150 and 300 K (REF, Student- t), so the analog columns stop at $b^* = 6, 5, 3, 2, 1$ and 0: 184, 154, 94, 65, 38 and 15 of 236 counters (Table T8; Figures T2A, T4B). CONS gives 94, 94, 38, 38, 15 and 0; PERM all 236 up to 48 K and 154 at 77 K. Column 0 qualifies at 300 K only because its counters carry weight 1.

The cell saving against an all-digital lane at the same temperature depends on accounting [E5] (QFA-3 at 12 cells against 15; four alignment cells per QFA in a mixed tree; full adders at 18 cells for wave-pipelined composition; the select stage in CMOS): -4 % to +25 % at 4 K, -3 % to +21 % at 20 K and -1 % to +9 % at 77 K (10.8 %, 9.0 % and 3.8 % in the record idiom: full adder 15 cells, QFA-3 12, select on fabric). The listing's all-digital reference (5,342–5,641 cells, with a re-sync allowance at Part I's 10–24-gate rule, which Section 2.2 retires, and no wave-pipeline balancing) is smaller than the timing-closed lane, so only ratios are used. Even if every counter were analog, the tiles would save at most about a third of the cells (≤ 32 %, with 18-cell full adders and the select stage in CMOS; 13.8 % in the record idiom), and only when cold.

6.5 Cooling economics

A cold lane pays for its cooler. The record's Joule model [E8], $P = n m^* v_0^2 / \tau$ at the tracked bias (n here the carrier density), scales as τ^{-3} and makes a cold cell one-eighth of a 300 K cell (0.97 μ W at and below 150 K, where the lifetime saturates), and it is uncertified below ≈ 255 K, where the kinetic closure of Part I §7.5 has not been evaluated (Part I §8.3; Part II §8). A cooler at efficiency η_c (a fraction of Carnot) spends $(T_a - T) / (\eta_c \cdot T)$ watts per watt removed — at Carnot 2.90 at 77 K, 14.0 at 20 K and 74.0 at 4 K from a 300 K ambient T_a — so a cold lane of N_{cold} cells draws, relative to the power P_{warm} of the warm all-digital lane of N_{warm} cells, a wall-plug power P_{wall} :

$$P_{wall}/P_{warm} = (N_{cold}/N_{warm}) \cdot r \cdot [1 + (T_a - T)/(\eta_c \cdot T)] \quad (T5)$$

where $r = P_{cell}(T)/P_{cell}(300 \text{ K})$ is the cold-to-warm ratio of cell dissipation. With $r = 0.125$ and an ideal cooler, the 77 K hybrid lane draws 0.47 and a 77 K all-digital lane 0.49 of the warm lane's power;

20 K and 4 K lanes draw 1.7 and 8.4 times and lose even at Carnot (Table T9; ternary_qmac_design.py §6). Break-even needs ≥ 39.6 % of Carnot for the 77 K hybrid and 41.4 % for the 77 K digital lane, and with equal cell dissipation ($r = 1$) every cold lane loses, by 3.8 to 67 times. Conversely, the hybrid breaks even at $r \leq 0.267$ with an ideal cooler, 0.126 at 40 % and 0.035 at 10 % of Carnot; the Joule model's 0.125 sits on the 40 % column (Figure T4C).

The analog counters thus move the 77 K lane's power by about 4 % (0.47 against 0.49); the rest is the decision to cool, which the digital lane can make equally — and under bookings A2–A3 would have to. Whether a cooler of the needed efficiency exists at the needed size is a question for the cooler, not the fabric [20].

Lane	Analog counters	Cells ÷ warm lane	Equal dissipation, ideal cooler	Joule model ($r = 0.125$), ideal cooler	Carnot fraction needed	Break-even r at 100 / 40 / 30 / 20 / 10 % of Carnot
TQ-4 (4 K)	184	0.892	66.9×	8.37×	beyond Carnot	—
TQ-20 (20 K)	154	0.910	13.6×	1.71×	beyond Carnot	—
TQ-77 (77 K)	65	0.962	3.75×	0.468×	39.6 %	0.267 / 0.126 / 0.098 / 0.067 / 0.035
Digital at 77 K	0	1	3.90×	0.487×	41.4 %	0.257 / 0.121 / 0.094 / 0.065 / 0.033

Table T9. Cooling economics (ternary_qmac_design.py §6; in-model, cold-side dissipation uncertified). Wall-plug power is relative to the warm (300 K) all-digital lane, with Carnot work per watt removed of 2.90 (77 K), 14.0 (20 K) and 74.0 (4 K) from a 300 K ambient. The cell ratios are 1 minus the record-idiom savings of Section 6.4 (10.8 %, 9.0 % and 3.8 %). Break-even r is the largest cold-to-warm cell-dissipation ratio at which the cold lane matches the warm one.

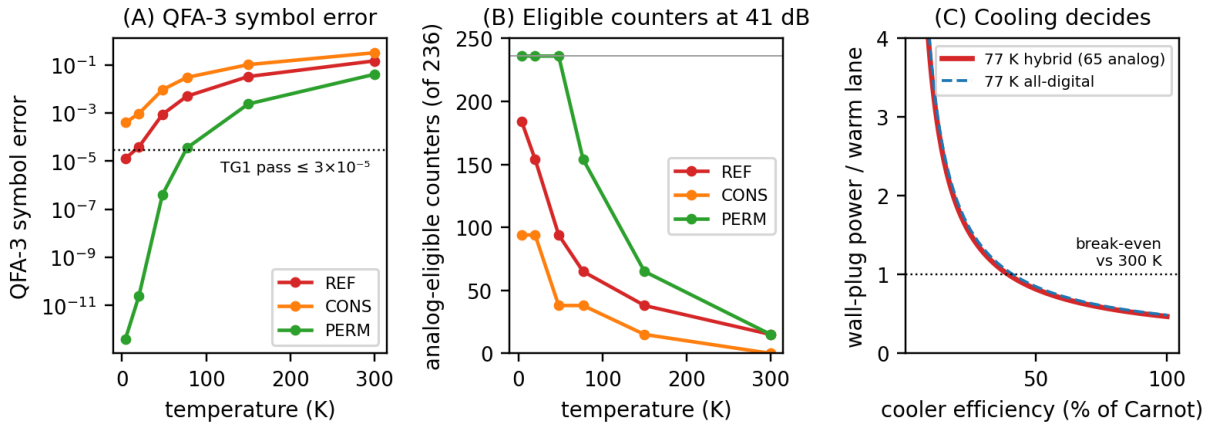


Figure T4. The analog counter tree on the Part-II fabric (Section 6; ternary_qmac_design.py §4–§6). (A) QFA-3 symbol error versus temperature under three conventions, marked by circles: REF (rail at the combiner output P2, relaunched inputs, $c = 1/\sqrt{12}$), CONS (REF with the rail at the pre-loss launch sum P1) and PERM (quiet launch, P2, multiport junctions, $c = 0.02$); the dotted line is gate TG1's pass threshold, 3×10^{-5} . (B) Counters of the 236-counter tree that can be QFA-3 tiles while the dot product keeps 41 dB, versus temperature, for the same conventions, with the same colours, markers and legend; the grey line is the full tree. (C) Wall-plug power of a 77 K lane divided by that of the warm (300 K) all-digital lane, versus cooler efficiency in percent of Carnot, under the record's Joule model (a cold cell dissipates 1/8 of a warm one): the hybrid lane (65 analog counters) solid, the all-digital lane dashed. The two coincide within a line width (0.468 against 0.487 at 100 % of Carnot) and cross the dotted break-even line near 40 % of Carnot. In-model; the cold-side cell dissipation is uncertified.

6.6 The booking fork

The hybrid lane of Section 6.4 books its digital columns at the released rate (booking A0 at 11 decisions per full adder: each half adder's five active cells and the OR; ternary_qmac_design.py §5 omits the four

carry-alignment buffers that digital_lane_bonsai.py §4 also flips). Under booking B — the calculus that prices QFA-3, applied to the digital half adders — the analog listing’s own all-digital lane (no balancing buffers) reaches 90.0 dB at 4 K, 50.8 dB at 48 K, 32.4 dB at 77 K and –4.2 dB at 300 K, meeting its 41-dB target only below 61 K (ternary_qmac_design.py §6); for the reference lane’s logic netlist (likewise without balancing buffers; an upper bound, Section 5.2), digital_lane_bonsai.py §4 gives –4.0 dB at 300 K, 32.6 dB at 77 K (mapped; its direct 77 K emulation prints 35.4 dB from ≈ 25 error events) and a 59 K pass temperature against the 42.5-dB pass line (Table T7). The two listings agree within 0.3 dB (32.4 against 32.6 dB at 77 K), and the verdict does not move. The listing does not recompute the hybrid under B; under B the digital lane itself must run below about 60 K, so the comparison is again one between two cold lanes and their coolers (Section 6.5), its numbers not computed.

7. The decision, and the measurements that would change it

Keep the digital lane as the path for Bonsai: exact as a netlist (Section 4.4), under 2 mm² for any memory up to 3 TB/s, and, given a launcher that meets DG4, dependent on one measurable quantity (DG1) to work warm. Keep QFA-3 as a test tile (gates TG1–TG5, beside Part II’s QG family): the smallest analog counter that removes a bit, pre-registered under three conventions, testing relaunch noise, the rail plane and multi-input linearity. It is not a route to a better Bonsai lane: it saves at most about a third of the cells, only when cold, and the cooler then decides.

Both routes end at one measurement, the loaded decision error of a cell, which picks among bookings A0–A3 and B on one side and among REF, CONS and PERM on the other; the record cannot compute it further, because the swing’s reference plane and a loaded cell’s effective sharpness are open.

The gates (Table T10) are pre-registered in Part I’s discipline and stand behind the prerequisites G1 (a cell has gain), L0 (Part I’s linewidth data gate, Section 2.4) and QG1–QG3. DG1 is decisive: its threshold is the pass line, 6.3×10^{-9} per decision (6.3×10^{-10} with a 10-dB margin), and its targets are the five bookings, spanning seven orders of magnitude in lane error power at 300 K, so any result names the machine. DG2 tests what no earlier gate does — a balanced, re-synced cascade as deep as the critical path, at the slot rate, for which the record has no multi-cell transient; DG3 tests the gated slot, DG4 the launcher (8–13 dB quieter than coded, on hundreds of lines), DG5 the heat, and TG1–TG5 QFA-3 and the cold-cell assumption.

If DG1 lands at A1, the warm lane needs about 8 K of cooling below 300 K or about 2 dB more model tolerance (61 K or 12.4 dB at the 353 K bias), priced in Sections 6.5 and 8.2. At A2 it is a 144 K machine and at A3 a 114 K one (colder at –3 dB junctions), under the arithmetic of Section 6.5, and at B a cryogenic one. If G1 fails, nothing physical in this paper survives.

Gate	Observable	Pass threshold	Model target (in-model) / source
G1	single-cell per-pulse net gain at ≤ 80 °C (prerequisite)	$\geq +2.6$ dB	+8 dB streaming; +9.2 dB design-length rerun (Part I Table 7)
L0	sub-threshold cavity linewidth of a specular-contact cell, with contact-preparation contrast, temperature and drift sweeps (prerequisite)	near-model linewidth (Part I §10)	model half-linewidth 27.6 GHz; ≈ 3.2 GHz of excess half-linewidth per 1 % additive round-trip loss (Part I §7.5, §10; notes 07-18)
QG1	two-pulse interference visibility at a junction combiner (prerequisite)	≥ 90 %	≥ 99 % (Part II Table Q4)
QG2	readout noise floor versus temperature, 300 $\rightarrow$ 4 K (prerequisite)	departs from the classical $\propto T$ line by ≥ 40 % at 20 K; flattens below ≈ 15 K	knee at $T_Q \approx 48$ K (Part II Table Q4)
QG3	QMAC-1 2-bit symbol error at 77 K, $N_{op} \approx 400$ (prerequisite)	$\leq 10^{-2}$	3.1×10^{-3} at –1 dB junctions; 5.6×10^{-2} at –3 dB (Part II Table Q4)

Gate	Observable	Pass threshold	Model target (in-model) / source
DG1	half-adder decision error, loaded (fan-out 2, real junctions, rail-level drive), at 300 K, 353 K and the lowest temperature reached	$\leq 6.3 \times 10^{-9}$ per decision, an upper bound (§5.2); $\leq 6.3 \times 10^{-10}$ for a 10-dB margin	at 300 K: $A0 \ 3.6 \times 10^{-11} \cdot A1 \ 9.7 \times 10^{-9} \cdot A2 \ 3.6 \times 10^{-5} \cdot A3 \ 2.0 \times 10^{-4}$ per decision; B 5.5×10^{-3} per half-adder decode (mapped onto the per-decision axis in Figure T3A) (§5.3)
DG2	path-balanced, re-synced cascade ≥ 92 cells (7–9 stations) at the slot rate, with measured cell delay	open eye; per-decision error as DG1	the record has no multi-cell transient at the gated point (§4.3)
DG3	gated $F = 2$ operation at the operating temperature	≥ 63 GHz with the ‘0’-slot floor ≥ 10 dB below data (report the strict 11.8-dB result too)	69 GHz [63, 74] at the 353 K bias (note 07-31); at the bench’s ratio bias neither confirmed nor moved: 67 GHz robust, 71 GHz marginal under the round specification (notes 08-01) and a 0.31-dB fail at its fitted convergence order (note 08-02, re-submitted; in-model)
DG4	launcher amplitude jitter and line count (384–704 launch, 26–91 readout lines at 69 Gb/s)	launch SNR ≥ 28 –33 dB (jitter ≤ 2.3 –4.0 %)	record’s coded launch: 20 dB (§5.4)
DG5	lane temperature rise at 3.6 % fill	≤ 10 K	2.8–3.7 K (300 K); 5.8–6.1 K (353 K) (§4.5)
TG1	QFA-3 four-level symbol error, relaunched inputs, N_{op} at P2, 4 K	$\leq 3 \times 10^{-5}$	1.3×10^{-5} REF; 3.5×10^{-7} quiet; 4.0×10^{-4} at P1; 9.9×10^{-4} at –3 dB junctions (§6.3)
TG2	linearity of the 3-input combiner with its top level at N_{rail}	minimum level gap $\geq 0.9 \times$ nominal	equally spaced by construction in the linear model
TG3	‘0’-level output noise σ_0 and ‘1’ jitter σ_{rel} over $\geq 10^4$ slots	$\sigma_0 \leq 1.56$ % of the rail swing; $\sigma_{rel} \leq 1.33$ % of the launched ‘1’ amplitude ($S \geq 37.5$ dB) — each the condition for p within $2 \times$ of the quiet-launch 3.5×10^{-7} at 4 K	released rule $\sigma_0 = 3.56$ % (fails; REF assumes it); coded launch 20 dB = 10 % (fails)
TG4	two-stage cascade QFA-3 $\rightarrow$ relaunch $\rightarrow$ QFA-3	error within $[0.5, 2] \times (p_1 + p_2)$	independent errors add
TG5	cold-cell dissipation ratio r (77 K) at the tracked bias, and cooler efficiency η_c	$r \leq$ the break-even entry of Table T9 at the measured η_c	Joule model $r = 0.125$, uncertified (§6.5)

Table T10. Pre-registered gates. G1, L0 and QG1–QG3 are prerequisites from Parts I and II; DG1–DG5 gate the digital lane and TG1–TG5 the QFA-3 test tile. DG1 is the decisive measurement. TG1’s threshold is among the open rework items of the analog listing (Appendix TA).

8. Alternatives and fine-tunings

This section asks what else Part III could have done with each construct it borrows (Section 2.5), and what the model side could do for the fabric. The numbers come from variants.py, extensions [V1]–[V5] (Tables T11, T12; Figure T5B), all in-model and none in the record.

8.1 Fabric-side alternatives

(a) A native threshold full adder [V2]. The record’s full adder is two half adders and an OR (15 cells, depth 7), yet the cell’s primitive is a threshold on summed fields, and a full adder is two thresholds: the carry is the majority of the three inputs (one cell, all three enhancing, high threshold), and the sum is one cell with the three inputs enhancing and a weight-2 depleting coupling from the carry, firing when $i_1 + i_2 + c_{in} - 2 \cdot \text{carry} \geq 1$. With three alignment buffers that is 5 cells of depth 2, and lane logic falls from 5,160 to 2,680 cells. Part I §5 supplies the ingredients but not their combination: a three-input fan-in and a weighted inhibitory input are not in the record, and the sharpness under a three-way sum is untested. Open; the cheapest fabric-side lever if it works, and (not computed) it would shorten the critical path behind the balancing cost.

(b) The select stage in CMOS. Moving the select to the host cuts the lane to 1,240 logic cells with the threshold full adder, but every product bit must then be launched per lane — 8 bits per MAC with

nothing shared — against $2 + 8/L$ (3.14 at seven lanes). That trades cheap cells for the most expensive resource, lines (Section 4.6). Considered and not adopted.

(c) Block width. Summing 64 or 128 products per slot (the weight group is 128) takes 491 or 1,002 counters, so logic per MAC stays near 161–164 cells, while readout bits per MAC fall from 0.406 to 0.219 and 0.117 and host scaling from 2.40 to 1.20 and 0.60 GFLOP per token (variants.py §1; extension [V1]). But a block wider than Q8_0's 32 sums activations with different scales, so it needs a runtime change — one activation scale per wider block — whose effect on the yardstick is not computed. Open.

(d) Timing tolerance. A 4-cell skew tolerance would cut balancing buffers from 2,616 to 581 (lane 6,134–6,420 cells), a 14-cell one to 189 (5,742–6,028; digital_lane_bonsai.py §2), if the gated cell's unmodelled delay spread allows it (DG2). A register alternative — Part I §6.4's small fast registers, holding a waiting carry instead of regenerating it — is not in the released chains and is not priced.

(e) Batching and the two-clock menu. At batch 1 the lanes are 51–88 % busy; a batch of two or more fills them (Section 4.6). The roadmap leaves open a two-clock architecture — 4-ps slots as an $F = 1$ transport and re-timing layer over slower logic (ROADMAP WP6) — and note 2026-07-31 finds that a 91-GHz pulse survives as an $F = 1$ wire. Here the 0.25-THz chain stays the pulse source and logic runs at 69 GHz; nothing in the record prices a faster operand-transport layer.

(f) What cannot be done. Under Part I's 20-dB link budget and 6-dB margin, station spacing cannot exceed 13 cells (if the coded launch jitter is physical at the N_{op} plane, the same law gives 4.4–7.8 cells; note 2026-08-25 §3, open). A fixed lane cannot exploit weight sparsity, since every biased cell dissipates whatever its product. And nothing accumulates on the fabric across slots: the lossless coherent 16-slot combiner behind Part II's published averaging column is unbuildable (note 2026-07-16), and the buildable version accumulates as charge off the fabric — as the host does here.

Configuration	Counters (stages)	Adder / output bits	Logic cells: record · TFA · TFA + CMOS select	Launched bits per MAC, $L = 2 / 3 / 7$	Readout bits per MAC	Host scaling, GFLOP per token
$b = 8, W = 32$ (reference)	236 (8)	12 / 13	$5,160 \cdot 2,680 \cdot 1,240$	6.00 / 4.67 / 3.14	0.406	2.40
$b = 6, W = 32$	176 (8)	10 / 11	$3,974 \cdot 2,114 \cdot 930$	5.00 / 4.00 / 2.86	0.344	2.40
$b = 4, W = 32$	116 (8)	8 / 9	$2,532 \cdot 1,292 \cdot 620$	4.00 / 3.33 / 2.57	0.281	2.40
$b = 2, W = 32$	56 (8)	6 / 7	$1,218 \cdot 598 \cdot 310$	3.00 / 2.67 / 2.29	0.219	2.40
$b = 8, W = 64$	491 (10)	13 / 14	$10,440$ (163.1 per MAC) · — · —	6.00 / 4.67 / 3.14	0.219	1.20
$b = 8, W = 128$	1,002 (12)	14 / 15	$21,000$ (164.1 per MAC) · — · —	6.00 / 4.67 / 3.14	0.117	0.60

Table T11. The lane's size under the levers (variants.py §1; in-model, before timing closure). b = activation bits, W = block width, TFA = the threshold full adder [V2] (not in the record). With the select stage in CMOS, every product bit is launched per lane: 8 bits per MAC at $b = 8$. Readout bits per MAC are output bits divided by W ; launched bits per MAC ($2 + b/L$) do not depend on W .

8.2 Model-side fine-tunings

The model can also move toward the fabric, using the techniques that made it ternary. None has been tried on Bonsai; the numbers below are in-model compositions on synthetic activations unless labelled otherwise, and whether the model keeps its quality under any of them is external and open.

(a) Activation precision. Quantization-aware fine-tuning lowers activation bits — BitNet a4.8 [21] reports 4-bit activations for 1.58-bit models at near-parity; LLM-QAT [22] is the general quantization-aware method, and SmoothQuant [23] a training-free post-training alternative. At $b = 6, 4$ and 2 bits the tree has 176, 116 and 56 counters, lane logic 3,974, 2,532 and 1,218 cells, and seven lanes launch 2.86,

2.57 and 2.29 bits per MAC instead of 3.14 (Table T11). The larger effect is on the error budget, since the model's quantization error grows faster than the lane's error weights shrink: the yardstick [V3] falls to 30.4, 17.9 and 3.5 dB (Student-t), κ_b [V4] to 1.8×10^7 , 1.3×10^6 and 6.8×10^4 , and the ceiling rises to 9.5×10^{-8} , 1.3×10^{-6} and 1.3×10^{-5} per decision (variants.py §2). At 4 bits A1 passes throughout the scanned range (to 400 K, the scan top), A2 below 213 K and A3 below 168 K, against 291, 144 and 114 K at 8 bits in the same emulation (Table T12; 292 K with the reference lane's κ , Section 5).

(b) Error-aware fine-tuning. Analog accelerators train with injected noise so that networks tolerate their hardware's errors [24]. If Bonsai tolerated fabric error power 10 or 100 times its Q8_0 error [V5], the pass line would fall from 42.5 to 32.5 or 22.5 dB, and the pass temperatures would become 338 K or the whole scanned range to 400 K (A1), 168 or 200 K (A2), and 132 or 158 K (A3) (variants.py §3). The lane is unchanged; and since its errors are rare and large, the training noise would have to be the lane's own, not a Gaussian.

(c) Regrouping the weight scale. One FP16 scale per 32 weights instead of per 128 adds $16/32 - 16/128 = 0.375$ bits per weight to PTQ1_0's 1.75 bits per weight [3] — about +21 % weight traffic, and so about 18 % fewer tokens per second in a short-context, memory-bound decode (external arithmetic; no listing computes it). It buys nothing: each 32-wide block already lies inside one 128-weight group, so every block sum already takes one weight scale and one activation scale. Considered and rejected.

(d) Ternary activations. At $b = 2$ the lane is 1,218 cells and 56 counters, but the yardstick collapses to 3.5 dB: a different model, not a fine-tuning. Listed for completeness.

In sum, Part III keeps the record's full adder and select-on-fabric design, with its own generalization of Part I's DLY balancing, as its reference, and considers five fabric-side alternatives (noting three impossibilities) and four model-side ones: the threshold full adder and the skew tolerance are worth measuring; wider blocks wait on a runtime change; select-in-CMOS and the regrouped scale are rejected; and 4-bit activations and error-aware fine-tuning (in-model, on synthetic activations; whether the model keeps its quality under them is external and open) are the levers that move the pass temperatures.

Quantity	$b = 8$	$b = 6$	$b = 4$	$b = 2$	$b = 8, 10 \times$ tolerance	$b = 8, 100 \times$ tolerance
Pass line (yardstick, Student-t)	42.5 dB	30.4 dB	17.9 dB	3.5 dB	32.5 dB	22.5 dB
Yardstick, Gaussian	45.5 dB	33.1 dB	20.3 dB	3.6 dB	—	—
κ_b (int ² per block per unit BER)	2.93×10^8	1.80×10^7	1.26×10^6	6.83×10^4	2.93×10^8	2.93×10^8
Ceiling per decision (Student-t)	6.1×10^{-9}	9.5×10^{-8}	1.3×10^{-6}	1.3×10^{-5}	10× the $b = 8$ value	100× the $b = 8$ value
A0 pass temperature	394 K	whole scan (to 400 K)	whole scan (to 400 K)	whole scan (to 400 K)	whole scan (to 400 K)	whole scan (to 400 K)
A1 pass temperature	291 K	350 K	whole scan (to 400 K)	whole scan (to 400 K)	338 K	whole scan (to 400 K)
A2 pass temperature	144 K	173 K	213 K	267 K	168 K	200 K
A3 pass temperature	114 K	137 K	168 K	211 K	132 K	158 K

Table T12. The error budget under the model-side levers (variants.py §2–§3; in-model compositions on synthetic activations). At $b = 8$, variants.py's emulation ($1,024 \times 5,120$ rows at BER 2×10^{-5}) gives $\kappa = 2.93 \times 10^8$ and a 6.1×10^{-9} ceiling, within the 13 % spread of the reference lane's 2.8×10^8 and 6.3×10^{-9} . The tolerance columns assume the model tolerates fabric error power 10× or 100× its Q8_0 error, a fine-tuning target rather than a record quantity. "Whole scan" means the booking passes at every scanned temperature, up to the 400 K top of the scan (variants.py prints "above 400 K").

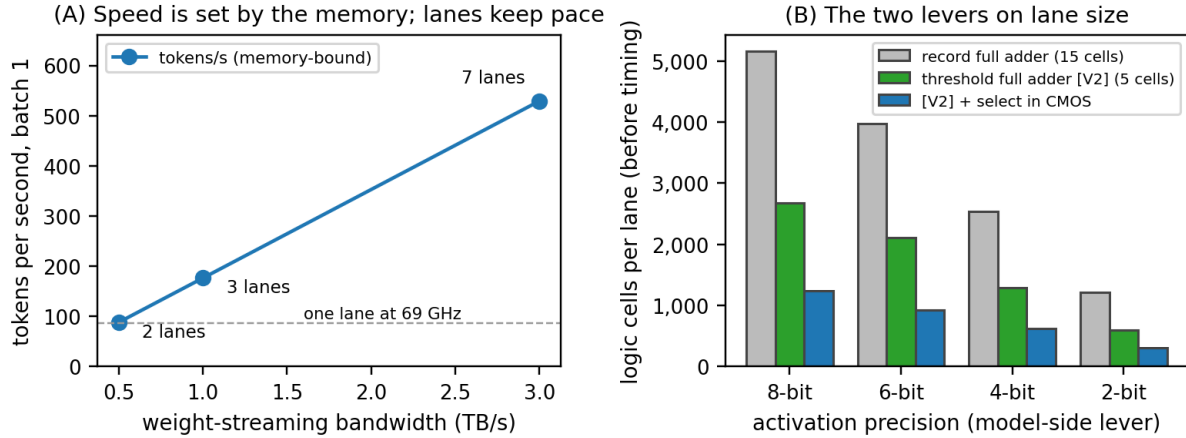


Figure T5. System scale and the levers (Sections 4.6 and 8). (A) Speed is set by the memory; lanes keep pace: tokens per second at batch 1 versus weight-streaming bandwidth, 88, 176 and 529 tok/s at 0.5, 1 and 3 TB/s (memory-bound, 5.669 GB per token), each point labelled beside it with the lanes needed to keep pace (2, 3 and 7); the dashed line is one lane at 69 GHz, 86.26 tok/s. (B) The two levers on lane size: logic cells per lane, before timing closure, versus activation precision (8, 6, 4 and 2 bits), for the record full adder (15 cells), the threshold full adder [V2] (5 cells) and [V2] with the select stage moved to CMOS — 5,160 / 2,680 / 1,240 cells at 8 bits down to 1,218 / 598 / 310 at 2 bits (variants.py §1). In-model; the threshold full adder is not in the record.

9. What is demonstrated, in-model, external and open; limitations

- **Demonstrated (runnable); netlist and arithmetic checks, not physics.** The lane netlist is exact on 1.70 M Bonsai-shaped block sums with bit-exact host scaling; the analog tree is exact on 20,000 random blocks; the K = 2 counter tile reproduces Part II's qmac.error_2bit bit for bit at 48 points; the workload count matches the file sizes; the record anchors are released values.
- **In-model.** Everything physical: timing-closed cell counts, power, energy, heat, bookings, pass temperatures, the launch requirement, QFA-3's envelope, eligible counters, cooling and every lever of Section 8.
- **External.** Bonsai's architecture, sizes, scores and decode rates [3–5, 10]; HBM bandwidth [18] and access energy [19]; the Q8_0 format [11], whose use for Bonsai is this paper's assumption (Section 3.1).
- **Open.** The cell's gain (G1) and the linewidth gate (L0) with the ± 1 dB contact allowance; the loaded decision error (DG1) and so the booking; the reference plane of a decision's swing and of the rail; the gated cascade and slot (DG2, DG3), including the floor specification and bias convention behind 69 GHz; the launcher (DG4); the activation format of PrismML's runtime (Q8_0 assumed); the model's tolerance of lane errors and of lower activation precision; QFA-3 and its noise (TG1–TG4); cold-cell dissipation (TG5).

The limitations, merged across the three design listings (digital_lane_bonsai.py, ternary_qmac_design.py and variants.py):

1. Everything physical is in-model and inherits an unproven gain cell; no cell, cascade, launcher or readout exists.
2. The lane netlist and depth model are this paper's [D1, D7]; cell delay is one unit everywhere (≈ 0.33 ps per cell, a third of the ≈ 1 ps that Part I's depth-3 half adder takes at its 4-ps baseline, not established at the 69-GHz gated point), with no spread.
3. There is no layout or routing: areas are cell counts $\times 27.4 \mu\text{m}^2$, without wires, guard cells, clock taps or line fan-in.

4. Bookings A1–A3 and B compose record pieces and are not record values; which applies depends on the open reference plane of a decision’s swing (notes 2026-07-15, 2026-07-17).
5. Decisions are modelled as independent; correlated errors, disorder, drift and metastability are not (a metastability sensitivity is carried for QFA-3 only).
6. κ is a Monte Carlo estimate (13 % spread) on the logic netlist, about 5,100 decisions per block; the timing-closed lane’s balancing buffers and stations are decisions too and are not emulated, so κ understates its error power by an uncomputed factor.
7. The 300 K power is a hydrodynamic-tier τ^{-3} scaling [D3] without the kinetic correction to the dissipation, which the record does not evaluate at 300 K; the 69-GHz slot is unmodelled at 300 K; and cold-side physics — lifetime saturation, the Joule model — is uncertified below the ≈ 255 K closure line.
8. Biased cells burn power when idle: lanes 51–88 % busy dissipate as if full, and the fJ-per-MAC figures assume full utilization.
9. The yardstick is a proxy; the model’s tolerance of lane errors is unmeasured.
10. The inputs are synthetic: Bonsai-shaped weights with an assumed one-third share of zeros, and Gaussian or Student-t activations rather than Bonsai’s own rotated ones; the Q8_0 activation format is itself an assumption (Section 3.1).
11. Memory, interface, launch and CMOS-perimeter energies are not booked; the wall-plug launch energy depends on a launch-coupling efficiency the record does not quantify (Part I §9.5).
12. The ≈ 800 lines at 69 Gb/s — modulation, phase and coupling into the acoustic plasmon — are not modelled, nor is a cryostat interface for them.
13. The rail plane decides analog feasibility and is unanchored: QFA-3’s 4 K error differs 30-fold between P1 and P2.
14. $N_{\text{op}} = 2,600$ is $6.5\times$ the registered 400, whose “launch-budget” cap the record never derives.
15. The relaunch model [E7] is minimal: a ‘1’ at the rail and a ‘0’ as the released amplifier’s unsaturated output, both attenuated to N_{op} .
16. Junction and split conventions [E1, E2] are choices, and the sensitivity to -3 dB junctions carries the second check’s caveat.
17. The comparator offset c is open: $1/\sqrt{12}$ in REF and CONS, 0.02 in PERM.
18. The launch-noise convention is Part II’s additive one at the N_{op} plane; the record’s four launcher conventions span $\times 239$ in decode error (note 2026-08-25, launcher-physics fork, check P3; Section 5.4).
19. The $k \geq 16$ comparator redesign is neither designed nor charged.
20. The booking fork (A0 or B) is unresolved; the analog verdict holds under both.
21. Cell-accounting variants are estimates, and the analog listing’s all-digital lane (5,342–5,641 cells) differs from the timing-closed lane (8,169–8,455).
22. The threshold full adder, its three-input fan-in and its weighted inhibitory coupling are untested.
23. The model-side levers are projections on synthetic data; no fine-tuning was run.
24. Not explored: phase-sensitive amplification or squeezing (Part II §7), a CMOS per-slot sampler, subranging decoders and wavelength-division or optical operand fan-out.
25. The 69-GHz slot rests on a floor specification and bias convention the record has not fixed (notes 2026-08-01, 2026-08-02).
26. The ± 1 dB contact allowance, possibly larger for a diffuse contact, is open and gated by L0 (Part I §7.5, §10).

10. Conclusion

Part I asked whether room-temperature graphene can gain back what a plasmon loses, and Part II what the fabric is worth when its pulses are counted in quanta. This Part asked what the fabric is for, given a demanding workload.

What maps is one inner loop, exactly. A 32-wide block sum of a ternary model's signed 8-bit products fits on a lane of Part-I cells that is exact by construction — demonstrated as a netlist on 1.70 M Bonsai-shaped block sums — and, in-model, takes 8,169–8,455 cells once timed for latch-free operation. What the fabric is for is energy and area at memory speed, not speed: two to seven lanes keep pace with 0.5–3 TB/s in at most about 1.7 mm² and 0.64 W at 300 K (hydrodynamic tier), the decode stays as fast as the memory and no faster, and the hardest engineering is the ≈800 operand lines that a fabric without storage needs.

What decides is one measurement. The lane is correct as physics only if a loaded cell decides as well as the released booking says; Part I's worst-path budget would make it a 144 K machine for the –3 dB split alone and a 114 K one through a –1 dB junction (colder at –3 dB), Part II's calculus a cryogenic one, and DG1 chooses. Interference counters help only when cold, save at most about a third of the cells and leave the comparison to the cooler: QFA-3 is a test tile, not a route. The levers lie mostly outside the fabric — 4-bit activations or error-aware fine-tuning would raise the pass temperatures by tens of kelvin, and a threshold full adder would halve the lane's logic if the cell supports a three-way sum.

None of the in-model statements is yet a measurement, and the gain cell beneath all of them is unproven. The design is now exactly as falsifiable as Parts I and II were built to be.

Appendix TA. The listings and their correction histories

The four listings live in fable-model-ternary/ (Table TA1). Each imports the released chains read-only (nothing edited, no bytecode written, run_all.py never run) and prints every number of its sections; the .out file beside each is its reference output (macOS arm64, Python 3.11.2, numpy 2.4.6; runtimes in Table TA1).

digital_lane_bonsai.py and ternary_qmac_design.py, the two design studies folded into Sections 3–7, are at their second revision (v2); before it, an independent adversarial verification agent re-checked each, and the findings were adopted. The digital-lane check found four material defects — unbooked wave-pipeline timing (now [D7]), the released booking shown only unloaded (now A1–A3), missing launch noise (now [D8]) and 300 K power paired with a slot rate adjudicated at 353 K (now stated), together with a superseded hydrodynamic 353 K low end (now kinetic-corrected) — and seven minor ones. The analog check found five — equal cold and warm cell dissipation (now [E8]), a misattributed rail plane (now P1 and P2, [E4]), extra junction levels without their quarter-wave, missing relaunched '0' noise (now [E7]) and uneven cell accounting — and seven minor ones. A second check of the analog listing raised five more: an unphysical gain at –3 dB junctions in the rail re-solve, the hybrid alignment accounting, a 1.7-dB SNR overstatement from per-block scales (Student-t; 0.2 dB for Gaussian), the TG1 threshold and the warm reference temperature. They are open rework items for a third revision that is on hold, and Section 6's numbers carry that caveat. The digital listing has since become v2.1, which also prints both ends of the heat range, the booking-B mapping factor (KAPPA_B_RATIO) and the mapped values; no computed quantity changed.

variants.py and figures.py were written for this manuscript. figures.py transcribes listing outputs, except that it computes the A-booking curves of Figure T3A from the released chains and maps the B points with the 300 K factor. The extensions each listing adds to the record, and the digital listing's external

inputs [D6], are defined one per line in Table TA3. The listings pre-register new result keys (Table TA2); none collides with either released results.json, and none contests a stored value.

File	Role	Extensions	Used in	Runtime
digital_lane_bonsai.py (v2.1)	the Part-I digital lane: workload, netlist, timing closure, bit-exact emulation, κ and the pass line, bookings A0–A3 and B with the booking-B mapping factor, launch noise, system, heat, CMOS perimeter	[D1]–[D8]	§2–8	21 s
ternary_qmac_design.py (v2)	the Part-II analog counter tree: K-input tile and its bit-for-bit check, split law, ternary encodings, QFA-3 envelope and sensitivities, hybrid lane, cooling, booking fork	[E1]–[E8]	§2, 4–7	2 s
variants.py	the levers: lane at b activation bits and block width W, threshold full adder, yardstick and ceiling at b bits, pass temperatures, tolerance	[V1]–[V5]	§8	16 s
figures.py	Figures T1–T5	—	all	—

Table TA1. The listings in *fable-model-ternary/*. Each prints every number of its sections; the .out files are the reference outputs on macOS arm64 (Python 3.11.2, numpy 2.4.6), and the runtimes are that machine’s.

Key	Value	Listing
dl_lane_cells_t0_low / _high	8,169 / 8,455	digital_lane_bonsai.py §6
dl_balancing_buffers_t0	2,616	digital_lane_bonsai.py §6
dl_kappa_int2_per_block_per_BER	2.82×10^8	digital_lane_bonsai.py §6
dl_ber_ceiling_equal_q80_t4	6.28×10^{-9}	digital_lane_bonsai.py §6
dl_snr_A0_300K_t4_dB / dl_snr_A1_300K_t4_dB	64.9 / 40.6	digital_lane_bonsai.py §6
dl_snr_B_300K_dB	−4.0	digital_lane_bonsai.py §6
dl_kappa_B_ratio_300K	0.051	digital_lane_bonsai.py §6
dl_launch_S_min_300K_k8_dB	27.9	digital_lane_bonsai.py §6
dl_lanes_for_3TBs	7	digital_lane_bonsai.py §6
qfa3_err_4K / 20K / 77K_p2_relaunch_c0289	1.302×10^{-5} / 3.917×10^{-5} / 5.016×10^{-3}	ternary_qmac_design.py §7
qfa3_err_4K_p1_relaunch_c0289	4.010×10^{-4}	ternary_qmac_design.py §7
qfa3_err_4K_p2_quiet_c002_multiport	4.058×10^{-13}	ternary_qmac_design.py §7
qfa3_N_op_p2_relaunch_cold	2,599.8	ternary_qmac_design.py §7
qfa3_T_below_1e-4_p2_relaunch_c0289_K	27.17	ternary_qmac_design.py §7
relaunch_zero_sigma_max_pct_4K	1.558	ternary_qmac_design.py §7
tq_allqfa_snr_dB_4K_p2_relaunch_t4	28.23	ternary_qmac_design.py §7
tq_bstar_77K_p2_relaunch_t4	2	ternary_qmac_design.py §7
tq77_carnot_fraction_needed_joule	0.396	ternary_qmac_design.py §7
ha_decode_error_bookingB_300K	5.479×10^{-3}	ternary_qmac_design.py §7
p3_counters_b8_W32	236	variants.py §4
p3_logic_cells_b8_W32_record / _b4_W32_record	5,160 / 2,532	variants.py §4
p3_logic_cells_b8_W32_TFA / _b4_W32_TFA_selectCMOS	2,680 / 620	variants.py §4
p3_ceiling_b8_t4 / p3_ceiling_b4_t4	6.05×10^{-9} / 1.25×10^{-6}	variants.py §4
p3_yardstick_b4_t4_dB	17.9	variants.py §4
p3_T_pass_A1_b8_K / p3_T_pass_A1_b4_K	291.3 / 400 (top of the scanned range)	variants.py §4
p3_T_pass_A2_tol10_K	167.6	variants.py §4

Table TA2. Pre-registered keys (new names, rounded to the precision the estimators support). None collides with *fable-model-chain/* or *fable-model-quantum/* results.json, and none contests a stored value.

Label	Definition (one sentence, from the listing header)	Listing
[D1]	The lane netlist: 32 int8 $\times$ ternary products per slot, select $(p \wedge a) \vee (n \wedge \neg a)$, sign bit inverted, a greedy Wallace tree of record full adders and a 12-bit ripple adder, with the block constants and scales applied in CMOS.	digital_lane_bonsai.py

Label	Definition (one sentence, from the listing header)	Listing
[D2]	Gate-level emulation in which every cell decision flips independently with probability BER.	digital_lane_bonsai.py
[D3]	τ^{-3} scaling of the stored 353 K cell dissipation to 300 K at the tracked bias (hydrodynamic tier).	digital_lane_bonsai.py
[D4]	Activation broadcast to L lanes through $F = 2$ buffer trees.	digital_lane_bonsai.py
[D5]	Bookings A1–A3 (the released margin law with the $F = 2$ softening, then the split and junction attenuation of Part I's worst path) and B (the Part-II calculus), beside the released booking A0.	digital_lane_bonsai.py
[D6]	The model's workload constants, external inputs from the public configuration and file sizes.	digital_lane_bonsai.py
[D7]	Timing: a depth model of the netlist in which every live wire is regenerated at every level (balancing buffers, skew tolerance t) and a re-sync station sits on every wire crossing a cut at every station spacing.	digital_lane_bonsai.py
[D8]	Launch noise: Part II's launch excess variance added at the first decision after each launch.	digital_lane_bonsai.py
[E1]	K-input combiner: a balanced binary tree of the released combiner, every arm passing the same extra junction levels, each with a quarter-wave of propagation.	ternary_qmac_design.py
[E2]	Comparator fan-out to K comparators: 3.0 dB per halving ($\log_2 K$) plus $\lceil \log_2 K \rceil$ junction levels, each with its quarter-wave; $K = 2$ is the released value exactly.	ternary_qmac_design.py
[E3]	Decision: released midpoint thresholds, worst-case symbol variance, decoder amplifier noise and the static offset term $V_{\text{static}} = c^2(x_{\text{range}}/k_{\text{dec}})^2$.	ternary_qmac_design.py
[E4]	Operating point: N_{op} capped by N_{rail} at a stated plane, P2 (top level at the combiner output) or P1 (the pre-loss launch sum).	ternary_qmac_design.py
[E5]	Cells: K comparators, thermometer-to-binary logic and alignment buffers in the record's idiom, with accounting variants carried explicitly.	ternary_qmac_design.py
[E6]	Signed tile: phase-encoded trit arms (phase $\pi = -1$, gated = 0) plus a coherent offset arm at combiner weight $\frac{1}{2}$.	ternary_qmac_design.py
[E7]	Relaunch: inside a tree a '1' is a rail pulse and a '0' a cell's unsaturated output, both attenuated to N_{op} before the released input transit.	ternary_qmac_design.py
[E8]	Cell dissipation by the released Joule form at the tracked bias, scaling as τ^{-3} ; uncertified below ≈ 255 K.	ternary_qmac_design.py
[V1]	The lane generalized to b activation bits and block width W in the same netlist idiom.	variants.py
[V2]	A native threshold full adder: carry as one majority cell, sum as one cell with a weight-2 depleting coupling from the carry, plus three alignment buffers (5 cells, depth 2).	variants.py
[V3]	The yardstick at b bits: the same absmax-per-32 quantizer at b bits (Q8_0 at $b = 8$), $K = 5,120$, Student-t and Gaussian activations, one-third zero weights.	variants.py
[V4]	κ_b by gate-level emulation with injected decision flips at b bits, and the per-decision ceiling that follows.	variants.py
[V5]	The tolerance lever: pass temperatures when the model tolerates fabric error power $1\times$, $10\times$ or $100\times$ its own quantization error.	variants.py

Table TA3. The extensions named in brackets throughout, one per line, as defined in the listing headers; [D6] is an external input, listed for completeness, not an in-model extension. Everything not labelled here is a released call of the record's chains.

Appendix TB. Symbols

Symbol	Meaning	Value or range
w; a	ternary weight; int8 activation (Q8_0)	$\{-1, 0, +1\}$; 8 bits
p, n	weight bit planes $[w = +1]$, $[w = -1]$; p is also the QFA-3 symbol error (§6) and n the carrier density (§6.5)	one wire each
x_m; u; U	product bit m; unsigned product; block sum, eq. (T1)	$u \in [0, 255]$; $0 \leq U \leq 8,160$
n_neg	number of -1 weights in a block	0–32
i ₁ , i ₂ , c_in; s, c_out	full-adder input bits and carry in; sum and carry out, eq. (T2)	bits
b; W	activation bits; block width	8; 32 (reference)
L	lanes	2 / 3 / 7
K	dot-product length (§5); counter inputs (§6)	5,120; 3 for QFA-3
N_B	blocks per K = 5,120 dot product	160
κ , κ_b	block-sum error variance per unit decision BER (at b bits)	$2.8 \times 10^8 \text{ int}^2$ ($b = 8$)
SNR_q; P_sig	yardstick (Q8_0 dot-product SNR); signal power in lane units, eq. (T3)	42.5 dB (Student-t), 45.5 dB (Gaussian)
BER_ceil	pass line per cell decision, eq. (T3)	6.3×10^{-9} (2.6×10^{-6} per launched bit)

Symbol	Meaning	Value or range
$k; k_{\text{eff}}$	threshold sharpness $\beta \cdot A_{\text{swing}}/\Delta n$ (Part I); at fan-out 2	8; $8/\sqrt{2} = 5.66$
F	fan-out	2
NF	cell noise figure (Part I's F)	2.77-dB floor; 4 dB
S; σ_{rel}	launch SNR; launch amplitude jitter (fraction of swing)	coded 20 dB = 10 %; needed 28–33 dB
$N_{\text{op}}; N_{\text{rail}}$	quanta per input pulse; rail quanta	2,600 (REF), 1,278 (CONS), 1,822 (PERM), 400 (registered); 3,833
$\bar{n}; T_{\text{Q}}$	thermal occupation of the mode; quantum crossover $\hbar\omega_0/k_{\text{B}}$	5.8 at 300 K; 48 K
c	comparator rms trim offset, fraction of transition width	$1/\sqrt{12}$ (REF, CONS); 0.02 (PERM)
σ_0	noise of a '0' level (relaunched), fraction of the rail swing	3.56 % released at 4 K; ≤ 1.56 % needed
P1, P2	rail reference planes: pre-loss launch sum; combiner output	CONS; REF, PERM
$\Delta; \eta_{\text{in}}, \eta_{\text{extra}}, \eta_{\text{fo}}$	per-comparator level spacing; transmissions, eq. (T4)	$\Delta^2 \propto K^{-3}$ (roughly) at P2
p; b*; j	QFA-3 symbol error (p is also a weight bit plane, §4); highest analog bit column; bit column	1.3×10^{-5} at 4 K (REF); 6 at 4 K
r; $\eta_{\text{c}}; T_{\text{a}}$	cold/warm cell dissipation ratio; cooler efficiency (fraction of Carnot); ambient, eq. (T5)	0.125 (Joule model); —; 300 K
$P_{\text{wall}}, P_{\text{warm}}; N_{\text{cold}}, N_{\text{warm}}; P_{\text{cell}}(T)$	wall-plug power of a cold lane and power of the warm all-digital lane; their cell counts; dissipation per cell at T, eq. (T5)	$P_{\text{wall}}/P_{\text{warm}}$ 0.47–8.4 (Joule model, ideal cooler); P_{cell} 7.78–10.22 μW (300 K), 0.97 μW (≤ 150 K)
$n; m^*; v_0$	carrier density; effective mass $E_{\text{F}}/v_{\text{F}}^2$; drift velocity (Part I; Joule form, §6.5)	10^{12} cm^{-2} ; —; $\approx 2.4 \times 10^5 \text{ m/s}$
M, M_{th}	drift Mach number; its DS threshold (Part I)	$M/M_{\text{th}} = 0.7$ (tracked)
τ	plasmon momentum-relaxation lifetime	1.0 ps (300 K), 0.85 ps (353 K), 2 ps saturated (≤ 150 K)

Table TB1. Symbols of Part III. Part-I symbols M , M_{th} , τ , k ($= \beta \cdot A_{\text{swing}}/\Delta n$), n (carrier density), m^* and v_0 carry over. Part I's plasmon speed s and cell length L are not used (s here is the full-adder sum, L the number of lanes), and Part I's noise figure F is written NF, since F denotes fan-out. K , p and n each carry two meanings, kept apart by section.

Data and code availability

The four listings are released in fable-model-ternary/ of the repository of Parts I and II (<https://github.com/ryoji-info/FableComputer>), beside the released chains they import read-only; this revision and the directory enter the repository together. Every computed number reproduces from the four scripts (Python 3 and numpy; matplotlib for the figures), with the reference outputs beside them; external and Part I/II values carry their citations, and in-text derivations are shown where they appear. The revision history is kept in papers/REVISION-HISTORY.md.

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